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GUIDELINES FOR MARINE LIFTING & LOWERING OPERATIONS

0027/ND

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PREFACE

This document has been drawn with care to address what are considered to be the primary issues in relation to the contents based on the experience of the GL Noble Denton Group of Companies ("the Group"). This should not, however, be taken to mean that this document deals comprehensively with all of the issues which will need to be addressed or even, where a particular matter is addressed, that this document sets out a definitive view for all situations. In using this document, it should be treated as giving guidelines for sound and prudent practice, but guidelines must be reviewed in each particular case by the responsible organisation in each project to ensure that the particular circumstances of that project are addressed in a way which is adequate and appropriate to ensure that the overall guidance given is sound and comprehensive.

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This document must be read in its entirety and is subject to any assumptions and qualifications expressed therein as well as in any other relevant communications by the Group in connection with it. Elements of this document contain detailed technical data which is intended for analysis only by persons possessing requisite expertise in its subject matter.

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1 SUMMARY

- 1.1 These guidelines have been developed for the design and approval of marine lifting operations, including subsea installations (but excluding pipelines and flowlines).
- 1.2 This document supersedes the previous revision, document No. 0027/ND Rev 9 dated 31 March 2010. The changes are described in Section 2.13.
- 1.3 These guidelines cover lifting operations by floating crane vessels, including crane barges, crane ships, semi-submersible crane vessels and jack-up crane vessels. They also include subsea installations using a crane, winch or derrick. They may also be applied to lifting operations by land-based cranes for the purpose of load-out. They are intended to lead to an approval by GL Noble Denton, which may be required where an operation is the subject of an insurance warranty, or where an independent third party review is required.
- 1.4 A description of the approval process is given for those projects which are the subject of an insurance warranty.
- 1.5 The report includes guidelines for the load and safety factors to be applied at the design stage.
- 1.6 Comments on the practical aspects of the management of the operation are also offered.

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2 INTRODUCTION

- 2.1 This document provides guidelines on which the design and approval of marine lifting operations may be based.
- 2.2 It covers lifting operations by floating crane vessels, including crane barges, crane ships, semi-submersible crane vessels, jack-up crane vessels, winches or derricks. It refers to lifting operations inshore and offshore and to installation of subsea equipment excluding pipelines and flowlines which are covered in 0029/ND "Guidelines for Submarine Pipeline Installation", Ref. [4]. Reference is also made to lifting operations by land-based cranes for the purpose of load-out or load-in onto or from a barge or other transportation vessel.
- 2.3 The guidelines and calculation methods set out in this report represent the views of GL Noble Denton and are considered to be sound and in accordance with offshore industry practice. Operators should also consider national and local regulations, which may be more stringent.
- 2.4 The Report includes guidelines for the safety factors to be applied, comments on safe rigging practice and the information and documentation to be produced by others in order to obtain GL Noble Denton approval.
- 2.5 Revision 2 superseded and replaced the previous version, Revision 1, dated 11th August 1993. Principal changes in Revision 2 included:
- Reference to the ISO Draft Standard on weight control
 - Reserves specified on weights as calculated or measured according to the ISO/DIS
 - Limitations of GL Noble Denton Approval clarified
 - Changes to the required clearances on pipelines and other subsea assets
 - Addition to a section on heave-compensated lifts
 - Addition of a section on lifts using Dynamic Positioning.
- 2.6 Revision 3 superseded and replaced Revision 2, and includes additional clarification on safety factors for shackles, and testing and certification requirements.
- 2.7 Revision 4 superseded and replaced Revision 3, and includes:
- Changes to referenced documents (Sections 2.3 and References)
 - Some changes to definitions (Section 3)
 - Changes to Dynamic Amplification Factors, to eliminate discontinuities (Section 5.7)
 - Elimination of an anomaly in the definition of Hook Load (Section 5.3)
 - Inclusion of consideration of fibre slings (Sections 5.10, 5.15 and 12)
 - Elimination of an anomaly in the treatment of spreader bars and frames (Sections 5.16 and 7.5)
 - Modification of the flow chart (old Section 5.16)
 - Changes to the derivation of bumper and guide design forces (Section 10.3).
- 2.8 Revision 5 superseded and replaced Revision 4, and corrected typographical errors in Table 5-1.
- 2.9 Revision 6 superseded and replaces Revision 5, and made the following principal revisions:
- The Guideline refers as appropriate to other standards, including
 - ISO International Standard ISO2408 - Steel wire ropes for General Purposes – Characteristics, Ref. [9]
 - ISO International Standard ISO 7531 - Wire Rope slings for General Purposes - Characteristics and Specifications , Ref. [10]
 - Definitions in Section 3 were generally revised and expanded.
 - Section 4.1.2 added for the Certificate of Approval

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- Section 5 was re-ordered, Figure 5-1 revised, DAF's expanded to include submerged lifts, guidelines for 1 crane-2 hook lifts added, yaw factor for inshore lifts deleted, use of alternative codes added, minimum sling angles included
 - Old Section 11 (Underwater Lifting) moved into Section 5.7.7
 - Section 5.6.8 added for inshore lifts made by jack-up crane vessels.
 - Section 5.6.9 expanded to include weather forecast levels.
 - Section 5.8.5 added: SKL for multi hook lifts.
 - Table 5-3: consequence factors revised.
 - Section 5.12.6 added: sling eye design.
 - Sections 6.3.1 and 8.7 added.
 - Old Section 12 (Heave compensated lifts) moved to Section 6.3.1
 - Section 8.5 expanded to include trunnions and sling retainers.
 - Clearances in Section 9.4 generally updated and expanded.
 - Dimensional control requirements added to 10.3 and design requirements in Section 10.5.4.
 - Sections 9.2.6 - 9.2.8 added: bumper and guide clearances and dropped objects.
 - Limitation on number of chained shackles and shackle orientation added in Section 12.10.5.
 - Section 13 updated, showing requirements for sling certificates, doubled sling restrictions and requirements for wire/sling type.
 - Old Section 13 (Lifts using DP) moved to Sections 12.7.1 and 12.8.9.
 - Sections 12.8.7 and 12.8.8 amended for in field environmental condition monitoring.
 - Section 12.8.10 added for risk assessments and HAZOPs
 - General text changes and revisions made.
- 2.10 Revision 7 superseded and replaced Revision 6. The changes were the removal of "by Floating Crane Vessels" in the document title and a correction in Section 5.14.1.
- 2.11 Revision 8 superseded and replaced Revision 7. The change was a correction in Section 5.12.5.
- 2.12 Revision 9 superseded and replaced Revision 8. The changes were:
- Definitions (Barge, IACS, Insurance Warranty, NDT, Survey, Vessel, Surveyor, Weather-Restricted Operation, and Weather-Unrestricted Operations) in Section 3 revised.
 - Text modified in Section 4.1.2.
 - Weather forecast needs modified in Section 4.3.1.
 - Weight and CoG factor for piles added in Section 5.2.5.
 - CoG factor included for lifts not using a CoG envelope in Section 5.5.4.
 - DAF for lifts 100t to 1000t revised in Table 5-1.
 - Text added in Section 5.8.6 for 4 unequal slings in a single hook lift.
 - Factor for fibre rope sling splices included in Section 5.11.1.
 - Radius changed to diameter in Section 5.12.5.
 - Shackle MBL used instead of sling MBL in Section 5.14.2
 - Text amended in Sections 6.2.4, 8.4.1 and 12.10.h.
 - Clause added for tuggers attached to lift points in Section 7.4.3.
 - Clearances clarified in Sections 8.7.2 , 9.2.1.
 - Bumper force increased in Section 10.4.1.d.
 - Secondary bumper and guide forces added in Section 10.4.4.
 - Set down loads added in Section 10.4.2.

- IACS member certification added in Sections 12.1.1 and 12.6.1.
- Sling certificate validity added in (old) Section 12.6.3.
- Spreader bar/frame certification added in (old) Sections 12.6.6 and 12.6.7
- Reference [3] (0032/ND – Guidelines for Moorings) added.
- Reference [9] (LR Lifting Code) added.
- Mooring analysis requirements added to Sections 12.1.1 and 12.7.3 to 12.7.7.

2.13 This Revision 10 supersedes and replaces Revision 9. Major changes are marked with a line in the right hand margin and are:

- The installation of subsea equipment has been added, mainly in Section 11.
- Part of the Approval Process has been moved from Section 4 to Section 4 of 0001/ND “General Guidelines for Marine Projects”, Ref. [1].
- Various changes and new headings in Figure 5-1.
- Weight control in Section 5.2 now references Section 8 of 0001/ND, Ref. [1].
- Clarification of Rigging Geometry in Section 5.4 and Lift Point Loads in Section 5.5.
- Text to consider measuring slings over pins included in Section 5.8.1.
- Section 5.9.4 added for 2-hook load factors and Sections 5.10.2, 5.10.4 and 5.10.5 for 2-part sling factors.
- Minimum safety factor for synthetic (fibre) slings reduced from 4.75 to 4.0 in Section 5.13.3.
- Clarification of shackle safety factors in Section 5.14.2 and grommets in Section 5.15.6.
- Allowance is made for DAFs already included in certified capacity in Section 7.5.2.
- Section 7.6.1 now references 0001/ND, Ref. [1] for load factors for structural steel.
- SLS and ULS limit states are replaced with LS1 (gravity dominated) and LS2 (environmental load dominated) in Sections 7.6.2 and 10.5.4.
- Clarification of sling ovalisation in Section 8.2
- Extra details provide of lift point inspection added to Section 8.6
- Section 8.8 (lateral lift point load) relocated from Section 5.
- Section 9.2.10 added for reduced clearances around lifted objects.
- Clearances around mooring lines and anchors has been transferred from old Section 9.4 to 0032/ND, “Guidelines for Moorings”, Ref. [6].
- Consideration of relative motion for lifting onto floating structures is included in Section 10.2.3.
- Section 12.3 now references 0001/ND, Ref. [1], for Weather Restricted Operations and Metocean Reduction Factors.
- Amplification of requirements for removing seafastings and other secondary steel before lifting starting in Section 12.9.3 and moving the transport vessel in Section 12.9.4.
- Additional guidance on slings and shackles in Section 12.10.
- Ref. [11] changed in Section 12.10.11 and in the Reference section.
- Guidance for use of Lifting Tools added in Section 12.11 and colour coding in Section 12.12.
- Information required for approval has been moved from the old Section 13 to Appendix A and the criteria in that section has been moved to earlier sections in the document.

2.14 All GL Noble Denton Guidelines can be downloaded from:

http://www.gl-nobledenton.com/en/rules_guidelines.php

2.15 Please contact the Technical Policy Board Secretary at TPB@nobledenton.com with any queries or feedback.

3 DEFINITIONS & ABBREVIATIONS

3.1 Referenced definitions are underlined.

Term or Acronym	Definition
50/50 weight estimate	The value representing the median value in the probability distribution of weight
9-Part sling	A sling made from a single laid <u>sling</u> braided nine times with the single laid <u>sling eyes</u> forming each eye of the 9-part sling.
Added Mass	Added mass or virtual mass is the inertia added to a system because an accelerating or decelerating body must move some volume of surrounding water as it moves through it, since the object and fluid cannot occupy the same physical space simultaneously. This is normally calculated as Mass of the water displaced by the <u>structure</u> multiplied by the <u>added mass coefficient</u> .
Added Mass Coefficient	Non-dimensional coefficient dependant on the overall shape of the structure
Approval	The act, by the designated <u>GL Noble Denton</u> representative, of issuing a <u>Certificate of Approval</u>
ASD	Allowable Stress Design (effectively the same as <u>WSD</u>)
Barge	A non-propelled <u>vessel</u> commonly used to carry cargo or equipment. (For the purposes of this document, the term Barge can be considered to include Pontoon, Ship or <u>Vessel</u> where appropriate).
Bending Reduction Factor / E_B	The reduction factor applied to the breaking load of a rope or cable to take account of the reduction in strength caused by bending round a shackle, <u>trunnion</u> , diverter or crane hook.
Cable-laid sling	A cable made up of 6 ropes laid up <u>over</u> a core rope, as shown in IMCA guidance, Ref. [7], with terminations at each end.
Certificate of Approval	A formal document issued by <u>GL Noble Denton</u> stating that, in its judgement and opinion, all reasonable checks, preparations and precautions have been taken to keep risks within acceptable limits, and an operation may proceed.
CGBL / Calculated Grommet Breaking Load	The load at which a <u>grommet</u> will break, calculated in accordance with one of the methods shown in IMCA guidance, Ref. [7]
Competent person	Someone who has sufficient training and experience or knowledge and other qualities that allow them to assist you properly. The level of competence required will depend on the complexity of the situation and the particular help required.
Consequence Factor	A factor to ensure that main structural members, lift points and <u>spreader bars / frames</u> have an increased factor of safety (including lateral loads) related to the consequence of their failure.
Crane vessel	The vessel, ship or <u>barge</u> on which lifting equipment is mounted. For the purposes of this report it is considered to include: crane barge, crane ship, derrick barge, floating shear-leg, heavy lift vessel, semi-submersible crane vessel (SSCV) and jack-up crane vessel.

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Term or Acronym	Definition
CRBL / Calculated Rope Breaking Load	The load at which a cable laid rope will break, calculated in accordance with one of the methods shown in IMCA guidance, Ref. [7].
CSBL / Calculated Sling Breaking Load	The load at which a sling will break, calculated in accordance with one of the methods shown in IMCA guidance, Ref. [7]. The breaking load for a sling takes into account the E_T (<u>Termination Efficiency Factor</u>) and E_B (<u>Bending Reduction Factor</u>) if greater.
DAF / Dynamic Amplification Factor	The factor by which the <u>gross weight</u> is multiplied, to account for accelerations and impacts during the lifting operation
Determinate lift	A lift where the slinging arrangement is such that the sling loads are statically determinate, and are not significantly affected by minor differences in sling length or elasticity
DP	Dynamic Positioning or Dynamically Positioned
E_B	See <u>Bending Reduction Factor</u>
E_T	See <u>Termination Efficiency Factor</u>
FAT	Factory Acceptance Test
FMEA or FMECA	Failure Modes and Effects Analysis or Failure Modes, Effects and Criticality Analysis
FoS	Factor of Safety
FSE	Free Surface Effect
GL Noble Denton	The legal entity trading under the GL Noble Denton name which is contracted to carry out the scope of work and issues a <u>Certificate of Approval</u> , or provides advice, recommendations or designs as a consultancy service.
Grommet	A grommet is comprised of a single length of unit rope laid up 6 times over a core, as shown in IMCA guidance, Ref. [7], to form an endless loop
Gross Weight	The calculated or weighed weight of the <u>structure</u> to be lifted including a weight contingency factor and excluding lift rigging. See also <u>NTE weight</u> .
Hook Load (Dynamic)	Dynamic hook load is static hook load times DAF
Hook Load (Static)	The Hook Load is the <u>Gross Weight</u> or NTE weight plus the <u>rigging weight</u>
IACS	International Association of Classification Societies
Indeterminate lift	Any lift where the sling loads are not statically <u>determinate</u>
Insurance Warranty	A clause in the insurance policy for a particular venture, requiring the approval of a marine operation by a specified independent survey house.
LARS	Launch And Recovery System
LAT	Lowest Astronomical Tide
LBL	Long Baseline Array
Lift point	The connection between the <u>rigging</u> and the <u>structure</u> to be lifted. May include <u>padear</u> , <u>padeye</u> or <u>trunnion</u>

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Term or Acronym	Definition
Load-in	The transfer of a major assembly or a module from a <u>barge</u> , e.g. by horizontal movement or by lifting
Load-out	The transfer of a major assembly or a module onto a <u>barge</u> , e.g. by horizontal movement or by lifting
LS1 / Limit State 1	A design condition where the loading is gravity dominated; also used when the exclusions of Section 9.2.5 of 0001/ND, Ref. [1] apply.
LS2 / Limit State 2	A design condition where the loading is dominated by environmental / storm loads, e.g. at the 10- or 50-year return period level or, for weather-restricted operations, where a <u>Metoccean Reduction Factor</u> according to Section 7.3.3 of 0001/ND, Ref [1], is to be applied.
Matched pair of slings	A matched pair of slings is fabricated or designed so that the difference in length does not exceed 0.5d, where d is the nominal diameter of the sling or <u>grommet</u> . See Section 2.2 of IMCA, Ref. [7]
MBL / Minimum Breaking Load	The minimum allowable value of <u>breaking load</u> for a particular <u>sling</u> , <u>grommet</u> , wire or chain etc.
Mechanical Termination	A sling eye termination formed by use of a ferrule that is mechanically swaged onto the rope. See ISO, Ref. [9] and [10]
Metoccean Reduction Factor	The maximum ratio of the operational criteria / design criteria to allow for weather forecasting inaccuracies. See Table 7-3 of 0001/ND, Ref. [1].
MPI / Magnetic Particle Inspection	A Non-Destructive Testing (<u>NDT</u>) process for detecting surface and slightly subsurface discontinuities in ferroelectric materials such as iron
NDT / Non Destructive Testing	Ultrasonic scanning, magnetic particle inspection, eddy current inspection or radiographic imaging or similar. May include visual inspection.
Net weight	The calculated or weighed weight of a <u>structure</u> , with no contingency or weighing allowance
NTE weight / Not To Exceed weight	Sometimes used in projects to define the maximum possible weight of a structure, excluding lift rigging.
Operation Duration	The planned duration of the operation excluding a contingency period from the <u>Point of No Return</u> to a condition when the operations /structures can safely withstand a seasonal design storm (also termed “safe to safe” duration).
Operational reference period	The <u>Operation Duration</u> , including a contingency period
Padear	A lift point consisting of a central member, which may be of tubular or flat plate form, with horizontal <u>trunnions</u> round which a sling or <u>grommet</u> may be passed
Padeye	A lift point consisting essentially of a plate, reinforced by cheek plates if necessary, with a hole through which a shackle may be connected
PLEM	Pipe Line End Manifold
PLET	Pipe Line End Termination

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Term or Acronym	Definition
PNR / Point of No Return	The last point in time, or a geographical point along a route, at which an operation could be aborted and returned to a safe condition
RAO	Response Amplitude Operator
Rigging	The slings, shackles and other devices including spreaders used to connect the <u>structure</u> to be lifted to the crane
Rigging Weight	The total weight of <u>rigging</u> , including slings, shackles and <u>spreaders</u> , including contingency.
Rope	The unit rope from which a <u>cable laid sling</u> or <u>grommet</u> may be constructed, made from either 6 or 8 strands around a steel core, as indicated in ISO Refs. [9] & [10] and IMCA, Ref. [7]
ROV	Remote Operated Vehicle
Seafasteners	The system used to attach a <u>structure</u> to a <u>barge</u> or vessel for transportation
Single Laid Sling	A cable made up of 6 ropes laid up over a core rope, as shown in ISO, Ref. [9] and [10], with terminations each end.
SKL / Skew Load Factor	The factor by which the load on any lift point or pair of lift points and rigging is multiplied to account for sling length mis-match in a statically <u>indeterminate lift</u>
Slamming loads	Transient loads on the structure due to wave impact when lifting through the splash zone.
Sling breaking load	The breaking load of a <u>sling</u> , being the calculated breaking load reduced by <u>termination efficiency factor</u> or <u>bending reduction factor</u> as appropriate.
Sling eye	A loop at each end of a sling, either formed by a splice or mechanical termination
Splice	That length of <u>sling</u> where the rope is connected back into itself by tucking the tails of the unit ropes back through the main body of the rope, after forming the <u>sling eye</u>
Spreader bar (frame)	A spreader bar or frame is a <u>structure</u> designed to resist the compression forces induced by angled <u>slings</u> , by altering the line of action of the force on a lift point into a vertical plane. The <u>structure</u> shall also resist bending moments due to geometry and tolerances.
Structure	The object to be lifted
Submerged Weight	<u>Gross Weight</u> of the <u>Structure</u> minus the weight of displaced water.
Survey	Attendance and inspection by a <u>GL Noble Denton</u> representative. Other surveys which may be required for a marine operation, including suitability, dimensional, structural, navigational and Class surveys.
Surveyor	The <u>GL Noble Denton</u> representative carrying out a <u>survey</u> . An employee of a contractor or Classification Society performing, for instance, a suitability, dimensional, structural, navigational or Class survey.
SWL / Safe Working Load	SWL is a derated value of <u>WLL</u> , following an assessment by a <u>competent person</u> of the maximum static load the item can sustain under the conditions in which the item is being used.

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Term or Acronym	Definition
Termination efficiency factor / E_T	The factor by which the breaking load of a wire or cable is multiplied to take account of the reduction of breaking load caused by a <u>splice</u> or mechanical termination.
TMS	Tether Management System
Tonnes	Metric tonnes of 1,000 kg (approximately 2,204.6 lbs) are used throughout this document. The necessary conversions must be made for equipment rated in long tons (2,240 lbs, approximately 1,016 kg) or short tons (2,000 lbs, approximately 907 kg).
Trunnion	A lift point consisting of a horizontal tubular cantilever, round which a sling or <u>grommet</u> may be passed. An upending trunnion is used to rotate a <u>structure</u> from horizontal to vertical, or vice versa, and the trunnion forms a bearing round which the sling, <u>grommet</u> or another <u>structure</u> will rotate.
ULC / Ultimate Load Capacity	Ultimate load capacity of a wire sling, grommet, chain, shackle or similar is the certified minimum breaking load. The ULC of slings and grommets allows for good quality splices. Ultimate load capacity of a padeye, clench plate, delta plate or similar structure, is defined as the load which will cause general failure of the structure or its connection into the barge or other structure.
USBL	Ultra Short Baseline array
UT / Ultrasonic Testing	Detection of flaws or measurement of thickness by the use of ultrasonic pulse-waves through steel or some other materials.
Vessel	A marine craft designed for the purpose of transportation by sea or construction activities offshore. See <u>Barge</u>
Weather restricted operation	A marine operation which can be completed within the limits of an <u>operational reference period</u> with a favourable weather forecast (generally less than 72 hours), taking contingencies into account. The design environmental condition need not reflect the statistical extremes for the area and season. A suitable factor should be applied between the operational weather limits and the design weather conditions (see Section 7.3.3 of 0001/ND, Ref. [1])
Weather unrestricted operation	An operation with an <u>operational reference period</u> greater than the reliable limits of a favourable weather forecast (generally less than 72 hours). The design weather conditions must reflect the statistical extremes for the area and season. The design weather is typically a 10 year seasonal storm, but subject to Section 7.2.2 of 0001/ND, Ref. [1].
WLL / Working Load Limit	The maximum force which a product is authorized to sustain in general service when the rigging and connection arrangements are in accordance with the design. See <u>SWL</u> .
WSD	Working Stress Design (effectively the same as <u>ASD</u>)

4 THE APPROVAL PROCESS

4.1 GL NOBLE DENTON APPROVAL

4.1.1 Further information on the approval process appears in Section 4 of Ref. [1] (0001/ND “General Guidelines for Marine Projects”).

4.1.2 Approval may be given for such operations as:

- Installation of liftable jackets
- Hook-assisted installation of launched jackets
- Installation of templates and other sub-sea equipment
- Handling of piles
- Installation of decks, topsides modules, bridges and flare towers/booms
- Load-outs and Load-ins
- Transfer of items between a transport barge and the deck of a crane vessel.

4.1.3 Lifts may be by a variety of crane configurations, including single cranes, two cranes on a single vessel, two or more cranes on separate vessels, single crane multi-hook sheerleg vessels, cranes mounted on jack-up vessels, or by one or more land based cranes.

4.1.4 GL Noble Denton approval may be given for the operation, including reviews of marine and engineering calculations and procedures, and consideration of:

- The actual and forecast weather conditions
- The suitability and readiness of all equipment
- The behaviour of the lifting vessel
- Any site changes in procedures
- The general conduct of the preparations for the operation.

4.1.5 A Certificate of Approval for a lift covers the marine operations involved in the lift only and is issued at the Point of No Return, at the start of the lifting operation. An offshore lift is normally deemed to start when cutting of seafasteners starts, after the crane is connected and slings partly tensioned. In exceptional cases procedures may be accepted in which a pre-agreed number of seafasteners are to be removed before the Point of No Return, as described in Section 9.4 of 0030/ND, Ref. [5]. It is normally deemed to be completed when the lifted object is set down in its intended position, and the crane(s) has been disconnected. For completion of lifted load-outs see Section 4.3 of 0013/ND, Ref. [2].

4.2 SCOPE OF WORK LEADING TO AN APPROVAL

4.2.1 In order to issue Certificates of Approval, GL Noble Denton will typically require to consider, as applicable, the following topics:

- The strength of the structure to be lifted, including the strength of the lift points.
- The capacity of the crane, taking into account the radius at which the lift will take place, whether the crane will be fixed or revolving and whether any down-rating is required for operations in the design seastate.
- The capacity of the crane in the event that multiple hooks are used to suspend /upend a load.
- The rigging arrangement, including slings, shackles and any spreader frames or beams, and the certification of the rigging components.
- The stability of the crane vessel during the lift, especially in the case of a ballasting malfunction.
- The mooring arrangements for the crane vessel, as outlined in Section 4.3.
- DP audit documentation and FMEA analysis and DP procedures detailing positioning systems (see Section 13.8 of 0001/ND, Ref. [1].

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- The limiting design weather conditions proposed, and the anticipated behaviour of the crane vessel in those conditions.
- The arrangements for handling and mooring the transport barge or vessel alongside the crane vessel.
- The arrangements for cutting seafastings before lifting.
- The management structure for the operations and Management of Change procedures.
- ROV performance documentation.
- Risk assessments, HAZOP /HAZID studies involving key personnel of all relevant parties.
- Simultaneous Marine Operations (SIMOPS).

4.2.2 The information required in order to issue a Certificate of Approval is listed in Appendix A.

4.2.3 Technical studies leading to the issue of a Certificate of Approval may consist of:

- a. Reviews of specifications, procedures and calculations submitted by the client or his contractors, or
- b. Independent analyses carried out by GL Noble Denton to verify the feasibility of the proposals, or
- c. A combination of third party reviews and independent analyses.

4.3 APPROVAL OF MOORINGS

4.3.1 A lift may normally be considered a weather- restricted operation. Limiting weather conditions for the lift operation shall be defined, taking into account:

- the weather forecast reliability and frequency for the area
- the duration of the operation, including a suitable contingency period
- the exposure of the site
- the time required for any operations before or after the lift operation, including crane vessel and transport barge movements.
- currents on the lifting vessel/transport barge during the lift.
- currents on the lifted structure during lowering through the water column.

4.3.2 An approval of a lift will normally include the approval of the crane vessel and transport barge moorings in the limiting design weather conditions specified for the lifting operation. When operating alongside an offshore installation, procedures should be submitted which show that the crane vessel and transport barge can and will be removed to a safe distance when the weather conditions exceed a specified level. An approval of a lift does not include approval of the vessel moorings in extreme weather conditions.

4.3.3 Similarly, an approval of a lifted load-out will include the approval of the crane vessel and transport barge moorings at the load-out quay in the limiting design weather conditions specified for load-out. It does not necessarily include approval of the crane vessel and/or transport barge moorings in extreme weather conditions. Note that for approval of load-outs, reference should also be made to GL Noble Denton Report 0013/ND - Guidelines for Load-Outs, Ref. [2].

4.3.4 Additionally, and if specifically requested, GL Noble Denton will study and issue an approval of the moorings of the crane vessel or the transport barge, for a more extended period.

4.4 LIMITATION OF APPROVAL

4.4.1 See Section 4.5 of General Guidelines 0001/ND, Ref. [1].

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4.5 SURVEYS

4.5.1 Where GL Noble Denton approval is required the surveys shown in Table 4-1 will usually be needed:

Table 4-1 Typically Required Surveys

Survey	Time	Place
Sighting of inspection / test certificates or release notes for spreader bars, lift points and attachments	Before departure of structure from shore and before offshore lift (if after a lifted load-out)	GL Noble Denton / client's office and / or fabrication yard
Sighting of certificates and inspection reports for slings and shackles. Inspection of rigging and laydown and rigging tie-down / seafastening		
Inspection of securing of loose items inside module		Fabrication yard
Inspection of Survey & Positioning equipment on structure and on seabed	Before departure and start of marine operations	Fabrication yard and lift site
Suitability survey of crane / installation vessel, if required	Before start of marine operations	As available
Crane / installation vessel mooring activities		At lift site
Crane / installation vessel in field DP trials		
Inspection of preparations for lift and issue of Certificate of Approval		
	Immediately before cutting seafastening	

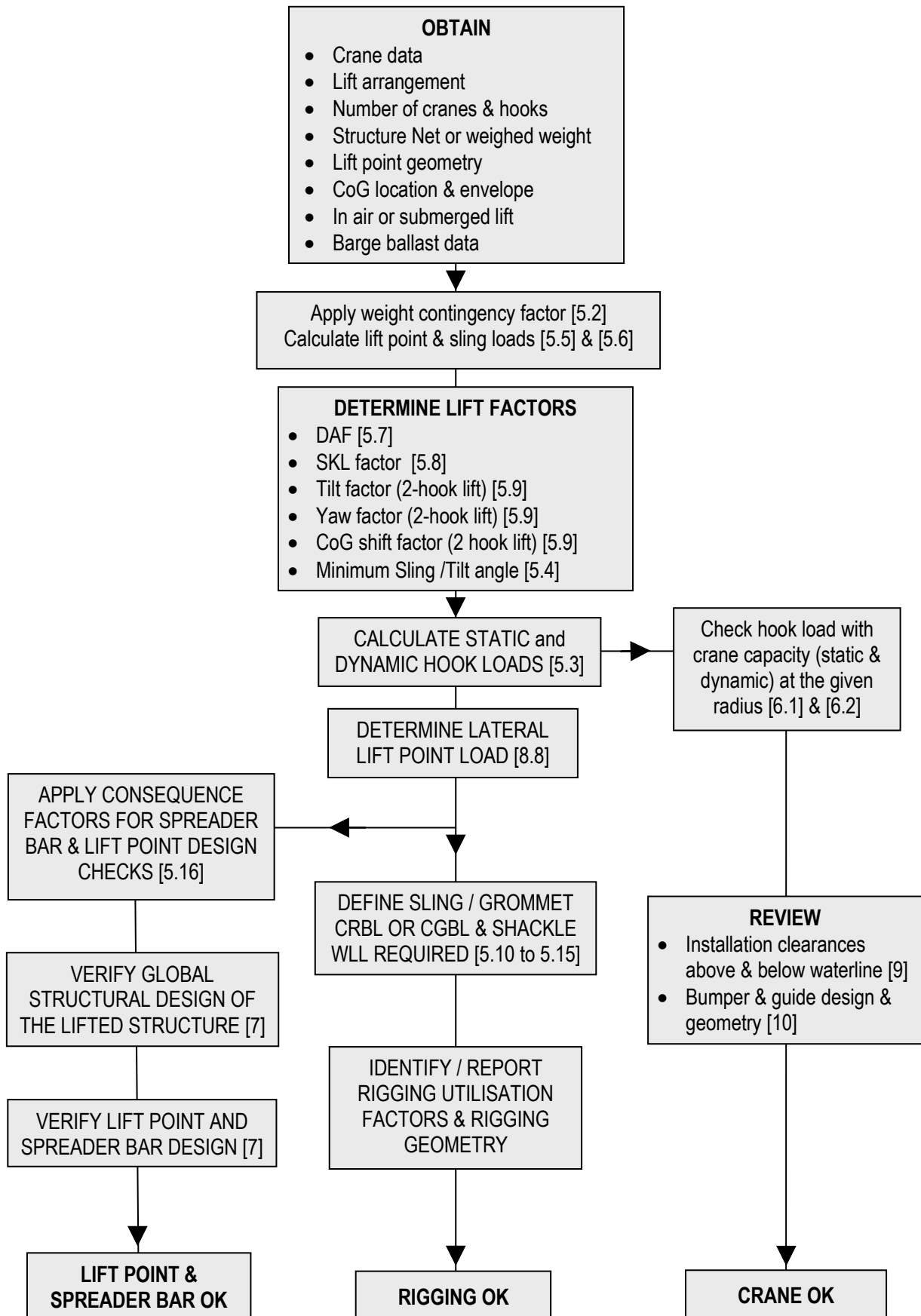
5 LOAD AND SAFETY FACTORS

5.1 INTRODUCTION

- 5.1.1 For any lift, the calculations carried out shall include allowances, safety factors, loads and load effects as described in these guidelines.
- 5.1.2 The various factors and their application are illustrated in Figure 5.1. This flowchart is for guidance only, and is not intended to cover every case. In case of any conflict between the flowchart and the text, the text shall govern. Figures in parentheses relate to sections in these guidelines.
- 5.1.3 Use of other recognised offshore codes of practice relating to lift engineering can also be considered, but care should be taken since not all other codes are exhaustive in determining the actual behaviour of lifting systems.

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Figure 5-1 Lift Calculation Flowchart



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5.2 WEIGHT CONTINGENCY FACTORS

- 5.2.1 Weight control requirements are given in Section 8 of 0001/ND, Ref. [1] which in turn references the ISO Standard 19901-5:2003, Ref. [8].
- 5.2.2 For Class A structure lifts (as defined by ISO Standard, Ref. [8]), the minimum weight contingency factor shall be 1.03 applied to the Net Weight for rigging and lift point design.
- 5.2.3 For Class B and C structure lifts (as defined by ISO Standard, Ref. [8]), the minimum weight contingency factor shall be 1.10 applied to the Net Weight for rigging and lift point design.
- 5.2.4 Except for piles, a weight contingency factor of not less than 1.03 shall generally be applied to the final weighed weight. This may be reduced if a Certificate is produced from a Competent Body stating, for the specific case in question, that the weighing accuracy is better than 3%.
- 5.2.5 The weight and CoG contingency factors for piles shall be determined considering the pile length and plate thickness tolerances.

5.3 HOOK LOADS

- 5.3.1 Loads in lift rigging and the total loading on the crane hook(s) should be based on hook loads defined as below, where:

$$\text{Static Hook load} = (\text{Gross Weight or NTE weight}) + (\text{Rigging Weight})$$

$$\text{Dynamic Hook load} = \text{Static Hook load} \times \text{DAF}$$

- 5.3.2 Rigging weight includes all items between the lift points and the crane hook, including slings, shackles and spreader bars or frames as appropriate.
- 5.3.3 For twin hook lifts whether cranes are on the same vessel, or multiple vessels, or the structure is suspended from two hooks on the same crane on the same vessel, the factors as described in Section 5.9 shall be used to derive individual hook loads.

5.4 MODULE TILT

- 5.4.1 The rigging geometry shall normally be configured so that the maximum tilt of the structure does not exceed 2 degrees. The sling angle should normally be as described in Section 5.6.2.
- 5.4.2 In special circumstances (e.g. flare booms, flare towers and cantilevered modules) the design angle of tilt may require to be greater than 2 degrees to permit the effective use of installation aids. These structures shall be reviewed as special cases.

5.5 LIFT POINT LOADS

- 5.5.1 The basic vertical lift point load is the load at a lift point, taking into account the structure Gross or NTE weight proportioned by the geometric distance of the centre of gravity from each of the lift points (if they are all at the same elevation). The basic lift point load is further increased by the factors as listed in Figure 5-1 as appropriate for the lifting arrangement under consideration.
- 5.5.2 If the lift points are at different elevations as shown in Figure 5-2 then sling forces shall be resolved at the sling intersection point, IP, which will be above the hook (if connected directly to the hook) or, if connected to a shackle /sling system suspended from the hook, the IP will be above the connection point on the shackle. The design sling loads should consider a CoG envelope and the loads in the slings determined by positioning the extremes of the CoG envelope under the IP and the sling loads recalculated using the new sling angles α and β .

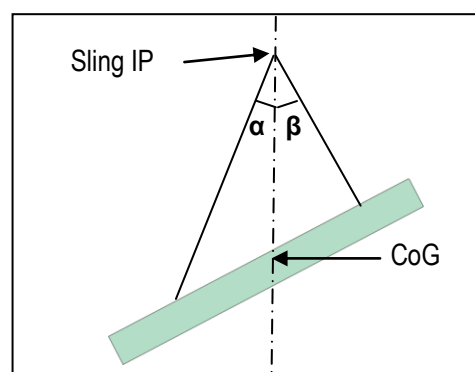


Figure 5-2 Resolving Sling Loading

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5.5.3 Where the allowable centre of gravity position is specified as a cruciform or other geometric envelope, then the most conservative centre of gravity position within the envelope should be taken. Where a CoG envelope is used, an additional factor of 1.03 should be added, to account for errors in the final CoG location from the weighing operation.

5.5.4 If a CoG envelope is not used then a CoG inaccuracy factor of 1.10 shall be applied to the weight.

5.5.5 For lift points where double trunnions or double padears are connected to a structure and are considered as a single lift point when determining loads, such as a double trunnion connected to the apex chord of a flare, the following effects of tilt and rotation shall be considered in the design of both structure and slings or grommets.

- Tilt can cause uneven loading unless there is means to ensure that the load on the two sides of the trunnion or padear is equalized.
- Tilt can also cause the rigging to shift along the bearing surface of the trunnion or padear such that increased moment is introduced into the trunnion or padear.
- As a result of friction, rotation of the sling eye or grommet round the padear or trunnion can result in significant torque on the padear or trunnion (and unequal loading in the legs of a grommet or doubled sling).

The use of a "matched pair" of slings or grommets connected to a double trunnion or double padear should be avoided as they are rarely adequately matched. If they are used, then the slings or grommets must have identical lengths when measured under the same tension. Where there are differences in the lengths, the effect of unequal lengths shall be considered in the design.

5.6 SLING LOADS

5.6.1 The sling load is the vertical lift point load resolved by the sling angle to determine the direct (axial) load in the sling and lift point using the minimum possible sling angle.

5.6.2 The sling angle should not normally be less than 60° to the horizontal although for lifts that are installed at an angle this may not be the case, e.g. flare booms installed by a single crane, the upper rigging may be less than 60°.

5.6.3 For lift point design, the rigging weight shall not form part of the lift point load.

5.6.4 For derivation of sling loads where the lift points are at different elevations, refer to Section 5.5.

5.7 DYNAMIC AMPLIFICATION FACTORS

5.7.1 Unless operation-specific calculations show otherwise, for lifts by a single crane in air, the DAF shall be derived from the following Table.

Table 5-1 Dynamic Amplification Factors (DAF) in Air

Gross weight, W (tonnes)	DAF			
	Offshore	Floating Inshore	Onshore	
			Moving	Static
$W \leq 100$	1.30	1.15		1.00
$100 < W \leq 500$	1.25	1.10		
$500 < W \leq 1,000$	1.20			
$1,000 < W \leq 2,500$	1.15	1.05		
$2,500 < W \leq 10,000$	1.10			

5.7.2 The DAF as indicated in Table 5-1 above shall also apply to the following in air lift combinations of vessels, cranes and locations:

- For lifts by 2 cranes on the same vessel
- For onshore lifts by 2 or more cranes

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- For lifts by 2 or more hooks on the same crane boom (but see Section 5.7.6 for offshore lifts)
 - For inshore lifts, in totally sheltered waters, by 2 or more vessels.
- 5.7.3 For lifts by cranes on jacked-up crane vessels:
- onto or from floating vessels use the “Offshore” or “Floating Inshore” column, as appropriate
 - onto fixed structures from its own deck, use the “Floating Inshore” column. If the crane is not moving horizontally on tracks or wheels, and horizontal motions of the load can be minimised by suitably located crane tuggers then the “Static” column may be used.
- 5.7.4 For onshore lifts, where the crane(s) may move horizontally, the “Moving” column of Table 5-1 shall apply. The “Static” column shall only apply if there is no crane movement other than lifting or lowering.
- 5.7.5 For offshore lifts by 2 or more vessels, the DAF shall be found by dynamic analysis.
- 5.7.6 For offshore lifts by 2 or more hooks on the same crane boom, total load on the crane boom structure shall be documented, based on Table 5-1 DAF’s increased by 1.10 unless certified crane curves for this specific application can be provided.
- 5.7.7 If any part of the lifting operation includes lifting or lowering a structure or spool through water, analyses shall be submitted, which either:
- Show how the total in-water lifting loads are derived, taking into account weight, buoyancy, entrained mass, boom-tip velocities and accelerations, inertia and drag forces, or;
 - Calculate the dynamic sling and hook loads to document that slack slings do not occur and provide limiting seastate data for the offshore operation.
 - Calculate the local and global stresses in the spool;
 - Calculate slamming loads on the structure being lifted.
 - The dynamic analysis results for a submerged or partially submerged lift may restrict the operability of an operation that is subject to the issue of a Certificate of Approval, depending on the DAF used for rigging and structure design.
- 5.7.8 As an alternative to the DAF’s in Table 5-1, the DAF may be derived from a suitable calculation or model test. Where the lift is from or onto a barge or vessel alongside the crane vessel, then the barge or vessel motions must be taken into account as well as the crane boom-tip motions.
- 5.8 SKEW LOAD FACTOR (SKL)**
- 5.8.1 Skew load is a load distribution factor based on
- sling length manufacturing tolerances, (see Section 2.2 of IMCA, Ref. [7])
 - sling measurement tolerances over measuring pins,
 - rigging arrangement and geometry,
 - fabrication tolerances for lift points,
 - sling elongation,
- and should be considered for any rigging arrangement and structure (see Section 7.2) that is not 100% determinate. A significantly higher SKL factor may be required for new slings used together with existing slings as one sling may exhibit more elongation than the others.
- 5.8.2 For indeterminate 4-sling lifts using matched pairs of slings, a Skew Load Factor (SKL) of 1.25 shall be applied to each diagonally opposite pair of lift points in turn. For unmatched slings the SKL shall be determined from detailed structural analysis.
- 5.8.3 For determinate lifts the SKL may be taken to be 1.0, provided it can be demonstrated that sling length errors do not significantly affect the load attitude or lift system geometry. The permitted length tolerance on matched pairs of slings is defined according to IMCA guidance, Ref. [7].
- 5.8.4 For a lift system using matched pairs of slings and incorporating:
- a. a single spreader bar a SKL of 1.05 is applicable.
 - b. more than one spreader bar a SKL of 1.10 is applicable.

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Note: for lifts where multiple hooks are used and each hook is connected to a single spreader bar, a SKL of 1.05 can be used.

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5.8.5 For multi hook lifts where the hook elevation can be shown to be individually controlled, a lower skew load factor than stated above may be applicable, subject to evaluation of sling length tolerances, rigging arrangement and crane operating procedures.

5.8.6 For a single hook lift where four slings of un-equal length are used (i.e. not Matched Pairs), the skew load factor shall be calculated by the designer (considering sling length tolerances and measured lengths) and applied to the structure and lift system design accordingly. Where the calculated SKL is less than 1.25 (as required in Section 5.8.2), an SKL of 1.25 shall be applied.

5.9 2-HOOK LIFT FACTORS

5.9.1 For a 2-hook lift (hooks on one or two cranes on the **same vessel**) the individual gross weight at each hook shall be multiplied by the following factors, to account for increased loads due to the tolerances of the elevation in the crane hooks:

Centre of gravity shift factor = 1.03

Tilt factor = 1.03

and if there are 2 slings to each hook, the load to each lift point shall be multiplied by a yaw factor, to account for tolerances in lift radii of the 2 hooks:

Yaw factor = 1.05

5.9.2 Factors reduced below those defined in Section 5.9.1 may be used, subject to supporting analyses, limiting seastate criteria and installation procedure steps/controls.

5.9.3 For 2 hook lifts where the crane hooks are located on **separate vessels** the factors in Section 5.9.1 shall be applied for **inshore** lifts, and be subject to calculation for **offshore** lifts.

5.9.4 For multi-hook lifts carried out by shear leg cranes (non-rotating crane), where the hook elevations are closely synchronised, the factors in Section 5.9.1 can be reduced by 50%. For shear leg type cranes on one vessel the yaw factor specified in Section 5.9.1 can be reduced to 1.0.

5.10 2-PART SLING FACTOR

5.10.1 Where a 2-part sling or grommet passes over, round or through a shackle, trunnion, padear or crane hook, other than at a termination, the total sling force shall be distributed into each part in the ratio 45:55 to account for frictional losses over the bend.

5.10.2 Where upending a structure requires the sling or grommet to slide over a trunnion or crane hook utilising a 2-part sling or grommet, other than at a termination, the total sling force shall be distributed into each part in the ratio 32.5:67.5 to account for frictional effects as the wire slides over the bend. For this condition, the ratio may be reduced if the lifting contractor can demonstrate through documented evidence or testing that a lesser value can be adopted.

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5.10.3 Where a 2-part sling or grommet passes over a rotating greased sheave on a trunnion the total sling force shall be distributed into each part in the ratio 49:51 to account for the frictional losses over the rotating sheave on the trunnion.

5.10.4 Where slings or grommets are used in any more than a 2-part configuration, calculations shall be submitted for review, and will require special consideration by GL Noble Denton.

5.10.5 When using fibre slings or grommets (i.e. Dyneema, HMPE, Round slings or webbing slings) in a doubled configuration the 2-part sling factor referenced in 5.10.1 shall be used for guidance, but the specific recommendations of the sling supplier should govern, based on the planned mode of use and the specifics of the sling type.

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5.11 TERMINATION EFFICIENCY FACTOR

5.11.1 The breaking load of a sling ending in a termination shall be the calculated rope breaking load multiplied by a factor as follows:

- For hand splices: 0.75
- For resin sockets: 1.00
- Swage fittings, e.g. "Superloop or Flemish Eye": 1.00
- Steel ferrules (mechanical termination): 0.80
- Fibre rope sling splices (Dyneema, HMPE): 0.90

Other methods of termination (i.e. 9-part slings) will require special consideration.

5.12 BENDING EFFICIENCY FACTOR

5.12.1 Where any wire rope sling or grommet is bent round a shackle, trunnion, padear or crane hook, the breaking load shall be assumed to be the calculated breaking load multiplied by a bending efficiency factor in accordance with IMCA guidance, Ref. [7]:

$$\text{Bending efficiency factor} = 1 - 0.5 / \sqrt{(D/d)},$$

where: **d** = the sling or cable laid rope diameter

D = the minimum diameter over which the sling body, sling eye, or grommet is bent.

5.12.2 For wire rope slings and grommets, this results in the bending efficiency factors detailed in the following Table 5-2.

Table 5-2 Bending Efficiency Factors

D/d	<1.0	1.0	1.5	2.0	3.0	4.0	5.0	6.0	7.0
Factor	Not allowable	0.50	0.59	0.65	0.71	0.75	0.78	0.80	0.81

5.12.3 For fibre rope slings, the bending efficiency may normally be taken as 0.9, provided the bending diameter is not less than the minimum specified by the manufacturer and subject to the specific recommendations of the sling manufacturer.

5.12.4 It should be noted that termination and bending factors should not be applied simultaneously. The one which results in the lower value of breaking load will govern, and should be used.

5.12.5 Under no circumstances should the sling or grommet body contact any surface where the diameter is less than 1.0d to maintain the sling in good condition under load. Bending in way of splices shall be avoided.

5.12.6 In certain circumstances, it will be necessary to check sling eye bending losses around a shackle or trunnion, where the D/d ratio is less than 4.0.

5.13 SLING OR GROMMET SAFETY FACTORS

5.13.1 The minimum safety factor on sling or grommet breaking load shall be calculated after:

- resolution of the load based on centre of gravity position and sling angle, and
- consideration of the factors shown in Sections 5.2 to 5.12 as appropriate.

5.13.2 For steel slings and grommets the minimum safety factor shall be not less than 2.25.

5.13.3 For fibre slings and grommets the minimum safety factor shall be not less than that recommended by the manufacturer or 4.0 if greater.

5.13.4 Further safety factors shall be applied to the sling design based on sling usage.

5.14 SHACKLE SAFETY FACTORS

- 5.14.1 The shackle WLL should not be less than the static sling load.
- 5.14.2 In addition to Section 5.14.1 above, the shackle MBL (WLL x safety factor) divided by a safety factor equal to 3.0 shall not be less than the dynamic sling load (static sling load x DAF).
- 5.14.3 Wide body shackles should not be connected bow-to-bow unless specifically allowed by the manufacturer.
- 5.14.4 Where the shackle is at the lower end of the rigging, the weight of the rigging components above the shackle, (including effects of the DAF and taking account of sling angle) may be deducted from the shackle load.

5.15 GROMMETS

- 5.15.1 Grommets require special consideration, to ensure that the rope breaking load and bending efficiency have been correctly taken into account. It is assumed that grommets are constructed and used in accordance with IMCA guidance, Ref. [7].
- 5.15.2 The load in a grommet shall be distributed into each part in the ratio 45:55, as indicated by Section 5.10.
- 5.15.3 The core of a grommet should be discounted when computing breaking load. The breaking load of a grommet is determined in accordance with IMCA guidance, Ref. [7].
- 5.15.4 The bending efficiency factors at each end of a grommet may differ, and the more severe value should be taken. Bending efficiency is derived as in Section 5.12 where rope diameter is the single part grommet diameter.
- 5.15.5 Bending in way of grommet butt and tuck positions shall be avoided. The location of the butt connection shall be marked.
- 5.15.6 When selecting a grommet, attention should be paid to the breaking load quoted by the supplier as this is **not** normally that for a single leg of the grommet but is the total for both legs of the grommet without bending reductions.

5.16 CONSEQUENCE FACTORS

- 5.16.1 The following consequence factors shall be further applied to the structure including lift points and the lateral load effects on lift points, and their attachments into the structure:

Table 5-3 Consequence Factors

Lift points including spreader bars, "strongbacks" and spreader frames	1.30
Attachments of lift points to structure or spool	1.30
Members directly supporting or framing into the lift points	1.15
Other structural members	1.00

- 5.16.2 The consequence factors shown in Table 5-3 shall be applied based on the calculated lift point loads after consideration of all the factors shown in Sections 5.2 through 5.10. If a partial load factor design is used then the consequence factors in Table 5-3 shall also be applied to the partial load factors for structural design. Consequence factors in Table 5-3 shall also be applied to lift point lateral loads.

5.17 FIBRE ROPE DEPLOYMENT SYSTEMS

- 5.17.1 Fibre rope deployment systems may be used for lowering structures to the seabed to reduce weight.
- 5.17.2 Elasticity and performance of fibre ropes used in a deployment system shall be provided by the manufacturer and be based on performance tests. The results of these tests shall be included in lift and lowering analyses.
- 5.17.3 The system shall be demonstrated to be adequate by load and function testing.
- 5.17.4 Certification for fibre ropes and the deployment systems shall be issued by an IACS member or other recognised certification body accepted by GL Noble Denton.

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- 5.17.5 Some systems/cranes use a combination of wire rope and fibre rope. Where fibre ropes or slings are attached to wire rope, the installation procedure shall clearly specify how these attachments are made.

6 THE CRANE AND INSTALLATION VESSEL

6.1 CRANES

- 6.1.1 If the crane is not certified by an IACS member, then an inspection by a competent person is required and his report demonstrating conformity shall be submitted.
- 6.1.2 A risk assessment shall be carried out in the presence of GL Noble Denton for cranes and lifting devices that are not a normal part of the vessel's equipment.

6.2 HOOK LOAD

- 6.2.1 The hook load shall be shown not to exceed the certified allowable crane capacity as taken from the load-radius curves. Crane curves are generally expressed as safe working loads or static capacities, in which case they should be compared against the dynamic hook load. Information should be obtained to document this.
- 6.2.2 The allowable load-radius curves as presented can sometimes include a dynamic effect allowance. If a suitable statement is received to this effect, then the dynamic hook load derived in Section 5.3 may be compared against the load-radius curves.
- 6.2.3 Some crane curves specify different allowable load curves for different seastates. These may similarly be taken to include dynamic effects. A seastate representing the probable limits for the operation should be chosen, and the static hook load used for comparison.
- 6.2.4 If the DAF included in the crane curves differs from the operation-specific value derived from Section 5.7, then the allowable static hook load should be adjusted accordingly but shall not exceed the certified crane (SWL or WLL) load-radius curve.

6.3 HEAVE COMPENSATION

- 6.3.1 Where heave compensated or constant tension lifts are planned, then the following information shall be obtained for the crane or cranes:
- Crane technical description and operating procedures,
 - Safe working load radius curves and boom slew angles in heave compensated mode or constant tension mode plus limiting seastates,
 - Crane de-rating curves,
 - FMEA for the crane system,
 - DAF analysis in heave compensated or constant tension modes,
 - Engine room/deck mechanics maintenance logs.
- 6.3.2 Additionally maximum and minimum crane loads for active heave or constant tension compensation should also be provided. Installation analyses should incorporate effects of heave compensation and demonstrate improved operability due to its activation.

6.4 INSTALLATION VESSEL

- 6.4.1 Installation vessels shall be capable of performing their intended functions within the seastates predicted for the job. The vessel deck layout shall be adequate for the equipment and provide sufficient clear personnel access ways.
- 6.4.2 In some cases it is likely that the structure will be loaded at an inshore location. In this case there must be adequate clearances around the structure when considering lifting operations, vessel movements and structure movements.
- 6.4.3 Some or all of the documentation and certificates in Section 6.6 of 0001/ND, Ref. [1] will be required, depending on the location of the operations and local regulations.

6.5 DP SYSTEMS (IF APPLICABLE)

- 6.5.1 A vessel with a minimum DP Class 2 will be required. However DP Class 3 vessels should be used in operations where the consequences of a loss of position are considered to have a reasonable potential to result in death, substantial structural damage or significant environmental pollution. See Section 13.2 of 0001/ND, Ref. [1] for more details
- 6.5.2 DP operating and positioning procedures (as applicable) should be documented and include station keeping analyses/rosettes, Vessel DP system FMEA & Annual Trials Reports etc as required in Section 13.8 of 0001/ND, Ref. [1].
- 6.5.3 For all lifts a minimum of 3 independent reference systems shall be provided.

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6.6 MOORING SYSTEMS (IF APPLICABLE)

- 6.6.1 The mooring arrangement for the operation and stand-off position shall be documented. This should include the lengths and specifications of all mooring wires and anchors, and a mooring plan showing adequate horizontal clearances on all platforms, pipelines and any other seabed obstructions. An elevation of the catenary for each mooring line, for upper and lower tension limits, shall demonstrate adequate vertical clearance over pipelines and horizontal clearance to fixed installations and the structure being lifted.
- 6.6.2 Anchor plans shall indicate the anchor position and the anchor line touch down point at the most likely working tensions. All subsea infrastructures and pipelines shall be shown together with exclusion zones. Anchor clearances to subsea assets shall be in accordance with see 0032/ND, Ref. [6].
- 6.6.3 Mooring analyses shall be submitted based on 0032/ND, Ref. [6], with calculations showing the anchor holding capacities for the soils expected, based on the site geotechnical data.
- 6.6.4 The mooring analysis shall also provide the limiting seastate for the installation vessel in the working position, including the transient motion due to a failed mooring line scenario.

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7 STRUCTURAL CALCULATIONS

7.1 CODES AND SPECIFICATIONS

- 7.1.1 For analysis of the structure to be lifted and the lift points, a recognised and applicable offshore structural design code shall be used as described in Section 7.6.
- 7.1.2 Adequate specifications for material properties, construction, welding, casting, inspection and testing shall be used.

7.2 LOAD CASES AND STRUCTURAL MODELLING

- 7.2.1 Structural calculations, based on the load factors discussed above, shall include adequate loadcases to justify the structure. For example, for an indeterminate, 4-point lift the following loadcases should normally be considered:
- Base case, using gross or NTE weight, resolved to the lift points, but with no skew load factor.
 - Gross or NTE weight, with skew load factor applied to one diagonal.
 - Gross or NTE weight, with skew load factor applied to the other diagonal.

- 7.2.2 In all cases the loading shall be applied at the correct or minimum sling angle and point of action, accounting for any offset. The effects of torsional loading imposed by the slings shall be considered.

7.3 STRUCTURE

- 7.3.1 The overall structure shall be analysed for the loadings shown in Section 7.2.
- 7.3.2 The primary supporting members shall be analysed using the most severe loading resulting from Section 7.1, with a consequence factor applied (see Section 5.16).

7.4 LIFT POINTS

- 7.4.1 An analysis of the lift points and attachments to the structure shall be performed, using the most severe load resulting from Section 7.2 and all the factors as appropriate from Section 5.
- 7.4.2 Where the lift point forms an integral part of the structural node, then the lift point calculations shall also include the effects of loads imposed by the members framing into the lift point.
- 7.4.3 Where tugger lines are attached to lift points their effect shall be considered in the lift point design.

7.5 SPREADER BARS, FRAMES & OTHER STRUCTURAL ITEMS OF LIFTING EQUIPMENT

- 7.5.1 Spreader bars, frames and other structural items of lifting equipment, if used, should be similarly treated, with loadcases as above. A consequence factor shall be applied to spreader bars and frames, in accordance with Section 5.16.
- 7.5.2 Where a spreader bar, frame or other structural item of lifting equipment is certified, the certified capacity may be increased by any DAF that is has been taken into account in the certified capacity before being compared against the dynamic loading enhanced by the applicable consequence factor from Section 5.16.

7.6 ALLOWABLE STRESSES

- 7.6.1 The primary structure shall be of high quality structural steelwork with full material certification and NDT inspection certificates showing appropriate levels of inspection. It shall be assessed using the methodology of a recognised and applicable offshore code including the associated load and resistance factors for LRFD codes or safety factors for ASD/WSD codes. Further details appear in Section 9.1 of 0001/ND, Ref. [1].
- 7.6.2 Except for sacrificial bumpers and guides, the loading shall be treated as an LS1 limit state.

7.7 INDEPENDENT ANALYSIS

- 7.7.1 Alternatively, GL Noble Denton will, if instructed, perform an independent analysis of the structure to be lifted, including the lift points, on receipt of the necessary information.

8 LIFT POINT DESIGN

8.1 INTRODUCTION

8.1.1 In addition to the structural requirements shown in Sections 5 and 7, the following should be taken into account in the lift point design.

8.2 SLING OVALISATION

8.2.1 Adequate clearance is required between cheek plates, shackle pins or inside trunnion keeper plates, to allow for sling ovalisation under load. In general, the width available for the sling shall be not less than $(1.25 \times d + 25\text{mm})$, where d is nominal sling diameter in mm. However, the practical aspects of the rigging and de-rigging operations may demand a greater clearance than this. The design of any lift point should account for the most onerous possible position of the sling on the lift point.

8.2.2 The bearing surface of cast padears in contact with the sling shall preferably be elliptical (rather than circular) in cross-sections orthogonal to the local sling axis, see Figure 8-1. This is to allow for the flattening/ovalisation of the sling cross-section which occurs when the sling is bent and under tension which allows more of the wire strands to work. It is not possible to generalise on dimensions for this purpose, as each sling would deform by a different amount, depending on the load that it is required to take. The sling arrangement should therefore be closely examined at an early stage in order to determine the amount of deformation which will need to be allowed for in the padear design.

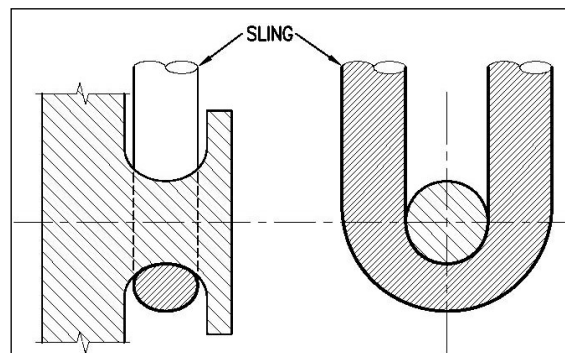


Figure 8-1 Indicative shaping of padear bearing surface

8.3 PLATE ROLLING AND LOADING DIRECTION

8.3.1 For fabricated lift points the direction of loading should generally be in line with the plate rolling direction. Lift point drawings should show the rolling direction.

8.3.2 Through-thickness loading of lift points and their attachments to the structure should be avoided if possible. If such loading cannot be avoided, the material used shall be documented to be free of laminations, with a recognised through-thickness designation.

8.4 PIN HOLES

8.4.1 Pin-holes should be bored /reamed, and should be designed to suit the shackle proposed. The pin hole diameter shall be 2 mm or 3% larger than the diameter of the shackle pin, whichever is the greater, up to a maximum of 6 mm.

8.5 CAST PADEARS AND WELDED TRUNNIONS

8.5.1 Cast padears and trunnions shall be designed taking into account the following aspects:

- The geometrical considerations as indicated in Section 8.2.
- The stress analysis and finite element design process (modelling and load application).
- Load paths, trunnion geometry and space and support for slings and grommets.
- The manufacturing process and quality control.
- Sling keeper plates shall be incorporated into the padear/trunnions design to prevent the loss of slings or grommets during load application and lifting. These devices shall be proportioned to allow easy rigging and de-rigging whilst being capable of supporting the weight of the sling section during transportation.

8.6 INSPECTION OF LIFT POINTS

- 8.6.1 All lift points shall be inspected prior to any lifting operation. For the first lift, the level of inspection for welds which form part of the load transfer system in the lift point shall be:
- 100% visual for all welds, and
 - 100% MPI (Magnetic Particle Inspection) for all fillet welds, partial penetration welds and butt welds, and
 - 100% UT (Ultrasonic Testing) for all butt welds.
- 8.6.2 For subsequent lifting operations using the same lift points, a visual inspection will be adequate provided that:
- The visual inspection shows no damage to the lift point material and welding
 - The rigging system is similar to that used for previous lifts and that the lift points have been designed for that rigging system
 - No excessive or uncontrolled loading has, or suspected to have, occurred during the preceding lifts
 - All design utilisations for primary load transfer in the lift points are less than 0.8.
- 8.6.3 Where the visual inspection shows that damage has occurred, appropriate repairs are to be taken which shall be subject to the 100% visual, MPI and UT requirements given in Section 8.6.1.
- 8.6.4 Where excessive or uncontrolled loading has occurred, and member /weld utilisations are less than 0.8, all primary welds are to be inspected with 100% visual and 50% MPI with butt welds also tested with 50% UT.
- 8.6.5 Where excessive or uncontrolled loading has occurred, and member /weld utilisations are greater than 0.8, all primary welds are to be inspected with 100% visual and 100% MPI with butt welds also tested with 100% UT.
- 8.6.6 Where the rigging system used is different to the design rigging system, all primary welds are to be inspected with 100% visual and 100% MPI with butt welds also tested with 100% UT.
- 8.6.7 For lift points with utilisations of primary load transfer members /welds greater than 0.8, the level of inspection shall be a minimum of 100% visual, 20% MPI with butt welds also tested with 20% UT.
- 8.6.8 The extent of NDT shall be submitted for review.

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8.7 CHEEK PLATES

- 8.7.1 Individual cheek plate thicknesses should not exceed 50% of the main plate thickness to maintain the primacy of the main plate in load transfer to the structure, and to provide robustness to lateral loads.
- 8.7.2 Non-load bearing spacer plates may be used to centralise shackle pins, by effectively increasing the padeye thickness. The diameter of the internal hole in such spacer plates shall be greater than the pin hole diameter. Spacer plates, if used, shall provide a 20-30 mm clearance to the inside width of the shackle (i.e. 10 to 15 mm each side).
- 8.7.3 Cheek plate welds shall be proportioned and designed with due regard to possible uneven bearing across the padeye/cheek plate thickness due to combined nominal (5%) and actual lateral loads.

8.8 LATERAL LIFT POINT LOAD

- 8.8.1 Provided the lift-point is correctly orientated with the sling direction, then a horizontal force equal to 5% of the resolved lift point load shall be applied, acting through the centreline and along the axis of the padeye pin-hole or trunnion /padear geometric centre.
- 8.8.2 If the lift point is not correctly orientated with the sling direction, then the computed forces acting transverse to the major lift point axis of the pin-hole or trunnion /padear geometric centre shall be added to the lateral lift point load as defined in Section 8.8.1. The effective length of the hook prongs shall be considered when finding the intersection point of the slings above the hook prongs.

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9 CLEARANCES

9.1 INTRODUCTION

- 9.1.1 The required clearances will depend on the nature of the lift, the proposed limiting weather conditions, the arrangement of bumpers and guides and the size and motion characteristics of the crane vessel and the transport barge.
- 9.1.2 Subject to the above, for offshore lifts, the following clearances should normally be maintained at each stage of the operation. Smaller clearances may be acceptable for inshore or onshore lifts. Clearances are based on a level lift (no tilt) of each structure. Additional clearances may be required for structures with a prescribed tilt.

9.2 CLEARANCES AROUND LIFTED OBJECT (FLOATING CRANE)

- 9.2.1 3 metres between any part of the lifted object (including spreaders and lift points) and the crane boom, when the load is suspended.
- 9.2.2 3 metres vertical clearance between the underside of the lifted object and any other previously installed structure, except in the immediate vicinity of the proposed landing area or installation aid where 1.5m clearance shall be adequate.
- 9.2.3 5 metres between the lifted object and other structures on the same transport barge unless bumpers and guides are used for lift-off.
- 9.2.4 3 metres horizontal clearance between the lifted object and any other previously installed structure, unless purpose-built guides or bumpers are fitted.
- 9.2.5 3 metres remaining travel between travelling block and fixed block at the maximum required load elevation with the lift vessel at LAT. "Remaining travel" excludes any travel prevented by system limits that cannot be over-ridden for this operation/
- 9.2.6 Where a structure is securely engaged within a bumper/guide or pin/bucket system, clearance between the extremities of the structure and the host structure must be demonstrated to be positive, considering the worst possible combinations of tilt. This may require dimensional control surveys to be carried out on the host structure and the structure to be installed.
- 9.2.7 Lift arrangement drawings shall clearly show all clearances as defined above.
- 9.2.8 Clearances for lifts by floating crane vessels onto floating structures (e.g. spars, FPSO's) will need special consideration. It is expected that the clearances for this case will need to be larger than those stated above. The design clearances will be dependent on the relative motions of the floating structure and the lifting vessel and should be agreed with GL Noble Denton.
- 9.2.9 Consideration should be given when lifting and overboarding structures over, or in the vicinity of, a subsea asset to provide sufficient horizontal clearance for dropped objects.
- 9.2.10 Clearances less than those stated above may be acceptable, but require special consideration and can result in reduced allowable metocean conditions for the lifting operation.

9.3 CLEARANCES AROUND LIFTED OBJECT (JACKED-UP CRANE)

- 9.3.1 The clearances in Section 9.2 above may be reduced to 1 metre for lifts to and from the jack-up deck (when elevated) and fixed structures provided that either suitable bumpers and guides or a very reliable manoeuvring system using tuggers or similar and an experienced crew are used. If bumpers and guides are used then the lifted object must be robust enough to withstand the likely loads from such aids without damage.

9.4 CLEARANCES AROUND CRANE VESSEL

9.4.1 Where the crane vessel is moored adjacent to an existing fixed platform with the mooring system intact the following clearances apply:

- 3m between any part of the crane vessel/crane and the fixed platform or lifted structure;
- 5m between any part of the crane vessel hull extremity and the submerged parts of the fixed platform or the submerged lift;
- 10 m between any anchor line and the fixed platform.

9.4.2 Where the crane vessel is dynamically positioned in accordance with DP Class 2 or 3, a 10 m nominal clearance between any part of the crane vessel and the fixed platform shall be maintained. This may be reduced in some cases to 5 m as described in Section 13.6 of 0001/ND, Ref. [1].

9.4.3 There should be a minimum **underkeel clearance** of 3m between crane vessel (including thrusters) and seabed, for an offshore lift after taking account of tidal conditions, vessel motions, increased draft and changes in heel or trim during the lift. Lesser clearance for operations in sheltered waters may be agreed with GL Noble Denton, depending on the seabed and environmental conditions, but should not be less than 1m.

9.4.4 Clearances around the crane vessel either moored or dynamically positioned and any floating platform, FPSO, drilling rig or submersible, shall be determined as special cases based on the station keeping analysis of the floating structure and the lifting vessel. Positioning equipment and procedures shall be defined to maintain the minimum clearances required for each specific operation. The procedures should minimise the durations for which these are required.

9.5 CLEARANCES AROUND MOORING LINES AND ANCHORS

9.5.1 The required clearances around mooring lines and anchors are given in Section 11 of 0032/ND "Guidelines for Moorings", Ref. [6]

9.5.2 Clearances should take into account the possible working and stand-off positions of the crane vessel.

9.5.3 During lifting operations, crossed mooring situations should be avoided wherever practical. Where crossed moorings cannot be avoided, the separation between active catenaries should be no less than 30 metres in water depths exceeding 100 metres, and 30% of water depth in water depths less than 100 metres.

9.5.4 If any of the clearances specified above are impractical because of the mooring configuration or seabed layout, a risk assessment shall be carried out and special precautions taken as necessary.

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10 BUMPERS AND GUIDES

10.1 INTRODUCTION

10.1.1 For module installation the arrangement and design philosophy for bumpers and guides shall be submitted, where applicable. In general, bumpers and guides should be designed in accordance with this Section taking into account of their use, configuration and geometry.

10.2 MODULE MOVEMENT

10.2.1 The maximum module movement relative to the target structure during installation should be defined. In general the relative motions should be limited to:

- Vertical movement: ± 0.75 m
- Horizontal movement: ± 1.50 m
- Longitudinal tilt: 2 degrees
- Transverse tilt: 2 degrees
- Plan rotation: 3 degrees.

10.2.2 The plan rotation limit is only applicable prior to engagement on the bumper/guide or pin/bucket system, and when the module is close to its final position or adjacent to another structure on a cargo barge.

10.2.3 Special consideration and agreement of relative motions with GL Noble Denton is required for cases where a module is being placed onto a floating target structure, as the motions of the target structure need to be considered.

10.3 POSITION OF BUMPERS AND GUIDES

10.3.1 The position of bumpers and guides shall be determined taking into account acceptable support points on the module.

10.3.2 Dimensional control reports for the as-built bumper and guide system shall be reviewed to ensure fit up offshore.

10.3.3 Nominal clearances between bumpers/guides and pins/buckets shall be ± 25 mm to account for fabrication and installation tolerances. These may be reduced based on trial fits and/or a stringent dimensional control regime.

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10.4 BUMPER AND GUIDE FORCES

10.4.1 For offshore lifts, bumpers and guides should be designed to the following forces (where W = static hook load) unless dynamic analyses are performed to justify alternative values:

Table 10-1 Default Bumper & Guide Forces (Offshore)

Lifting onto:	Fixed platform	Floating unit	Own deck
Vertical sliding bumpers			
Horizontal force in plane of bumper:	0.10 x W	0.20 x W	0.05 x W
Horizontal (friction) force, out of plane of bumper:	0.05 x W	0.10 x W	0.025 x W
Vertical (friction) force:	0.01 x W	0.20 x W	0.05 x W
Forces in all 3 directions shall be combined to establish the worst design case			
Pin/bucket guides			
Horizontal force on cone/end of pin	0.05 x W	0.10 x W	0.025 x W
Vertical force on cone/end of pin:	0.10 x W	0.20 x W	0.05 x W
Horizontal force in any direction shall be combined with the vertical force to establish the worst design case.			
Horizontal “cow-horn” type bumpers with vertical guide			
Horizontal force in any direction:	0.10 x W	0.20 x W	0.05 x W
Vertical (friction) force:	0.01 x W	0.20 x W	0.05 x W
Horizontal force in any direction shall be combined with the vertical force to establish the worst design case.			
Vertical “cow-horn” type guide with horizontal bumper			
Horizontal force in any direction:	0.10 x W	0.20 x W	0.05 x W
Vertical force on inclined guide-face:	0.10 x W	0.20 x W	0.05 x W
Horizontal force in any direction shall be combined with the vertical force to establish the worst design case.			

10.4.2 For inshore lifts under controlled conditions, bumpers and guides may be designed to 50% of the forces shown in Table 10-1.

10.4.3 Bumpers and guides that are deemed to arrest secondary forces (after the primary bumper and guide system has arrested the primary impact forces) may be designed to 50% of the forces shown in Table 10-1.

10.4.4 Lower bumper and guide forces may be agreed with GL Noble Denton for decommissioning structures when local damage may be acceptable.

10.5 DESIGN CONSIDERATIONS

10.5.1 The connection into the module, and the members framing the bumper or guide location, should be at least as strong as the bumper or guide.

10.5.2 The stiffness of bumper and guide members should be as low as possible, in order that they may deflect appreciably without yielding.

10.5.3 Design of bumpers and guides should cater for easy sliding motion of the guide in contact with a bumper. Sloping members should be at an acute angle to the vertical. Ledges and sharp corners should be avoided in areas of possible contact, and weld beads should be ground flush.

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- 10.5.4 With reference to Section 7.6, the strength of bumpers and guides that are deemed to be “sacrificial” may be assessed to the LS2 (environmental load dominated) limit state. The bumper and guide connection to the supporting structure shall be assessed to the LS1 limit state.

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11 INSTALLATION OF SUBSEA EQUIPMENT

11.1 SCOPE

11.1.1 These guidelines are specifically related to the installation of the following items subsea:

- Manifolds, Protection Structures, Templates and Production Equipment, including:
 - Xmas trees
 - Choke systems
 - BOP's
 - Subsea production equipment (multiphase pumps, separators, compressors, etc)
 - Control modules
 - Guide posts
 - Intervention tooling.
 - Ancillary equipment e.g. concrete mattresses, PLETs, PLEMs (when not installed as part of pipelay operations).
- Anchor piles (see Section 11.8 for suction and Section 11.9 for driven piles)
- Jumpers and tie-in spools (see Section 11.10)
- Rigid Pipe Risers (see Section 11.11)
- Storage Tanks (see Section 11.12)
- Tidal & wave generators

11.1.2 The installation of:

- Pipelines, flowlines and risers from a laying vessel, and
- PLETs & PLEMs when being installed as part of the pipeline/ pipe string

is covered by 0029/ND, Guidelines for Submarine Pipeline Installation, Ref. [4]. However, in some instances, the installation of a PLET at the initiation or termination of the pipelay will be covered by both guidelines; for example a PLET connected to line pipe but lowered by crane and winch with a damping system using buoys.

11.2 DESIGN PRINCIPLES

11.2.1 For any installation of subsea equipment, the calculations carried out shall include allowances, safety factors, load and load effects described in Sections 5 and 11.3.

11.2.2 The following expressions define the forces on the lifting systems:

Total force = Static force $\pm$ Hydrodynamic force

Static force = $m g - \rho V g$

Slam force (splash zone) = $0.5 \rho C_s A v_s^2$

Hydrodynamic force = $\sum$ slam, buoyancy, drag and inertia

Where:

m = object mass

ρ = density of seawater

g = the acceleration due to gravity

V = the volume of displaced seawater

v_s = the velocity of the object

A = the projected area of the object

C_s = the slamming coefficient. C_s may be taken as 3 for smooth cylinders and not less than 5 for all other objects.

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- 11.2.3 Rigging shock loads due to the submerged weight being exceeded by the hydrodynamic loads shall be avoided and where necessary the operational sea state shall be limited to ensure this. As a minimum, installation analyses should demonstrate that the minimum tension in the rigging is not less than 10% of its static value.
- 11.2.4 Lift / lowering analyses shall be carried out using appropriate software in the time domain to derive limiting seastates and their directions, the forces in the lifting system and the lifted item. Information on the software is required to evaluate its suitability.
- 11.2.5 The time domain simulation should be of sufficient duration to guarantee that the results are independent of the simulation time.
- 11.2.6 In lift / lowering analyses the calculation of boom tip motions shall be based on the actual vessel RAO and the corresponding offset of the lifting wire. The crane tip vertical motion shall be expressed as follows:

$$\text{Vertical boom tip motion} = \{H_m^2 + [t \sin(\theta_r)]^2 + [l \sin(\theta_p)]^2\}^{0.5}$$

Where:

H_m = the heave motion

t = the transverse offset of the lifting wire

θ_r = the roll motion

l = the longitudinal offset of the lifting wire

θ_p = the pitch motion

- 11.2.7 The dynamic amplification factor used for the design of the lifting system should be calculated as follows:

$$DAF = \frac{F_{\text{Static}} + F_{\text{Hydrodynamic}}}{F_{\text{Static}}}$$

- 11.2.8 The lift / lowering procedure and associated analyses should:
- highlight the operational steps to avoid/minimise slamming loads / remove likelihood of impact during splash zone lowering.
 - highlight the maximum allowable sea state and environmental conditions (in terms of wave height, wave period, wind speed and direction).
 - relate the maximum allowable environmental conditions to the maximum allowable tip motion of deployment system, and provide the maximum allowable tip motion.

When the tip motion is not readily available on the installation vessel the limiting heave, pitch and roll motion of the vessel shall be given (i.e. the heave, pitch and roll motions yielding the maximum allowable crane tip motion according to the installation analysis).

11.3 SUBSEA LIFTING REQUIREMENTS (ADDITIONAL TO THOSE IN AIR)

- 11.3.1 **Added Mass.** Hydrodynamic loads shall be calculated and take into account the added mass, drag, entrained water and buoyancy of the structure to be installed. Typically added mass will be considered for the lift and lowering analysis of a structure off a vessel deck, through the splash zone and water column and into position on the seabed or the host structure. Limiting sea states shall be defined based on results of installation analysis.
- 11.3.2 Calculations shall be undertaken to demonstrate that the total submerged weight of the lift including rigging is within the operating range of the proposed lifting / lowering system which may include heave compensation, cranes, winches and/or strand jacks.
- 11.3.3 The increased rigging loads due to entrained water and added mass shall be calculated and accounted for in the rigging design. Particular attention shall be applied where the lifted equipment is light compared to its rigging. Appropriate steps shall be taken to reduce the likelihood of the lifted equipment impacting its own rigging or of the rigging becoming slack at any time.
- 11.3.4 **Splash Zone.** When lifted equipment goes through the splash zone particular attention shall be paid to the slamming loads induced by prevailing seastate and/or vessel motion. The crane tip(s) shall be

arranged in such a position that the tip accelerations are kept to a minimum. Lifting or lowering through a vessel moon pool will normally increase the allowable seastate for operations. The local strength of the lifted equipment (including any hatch covers or ancillary equipment) should be checked for the effects of wave slamming.

- 11.3.5 **Tilt.** The weight and CoG of the lifted equipment in air and water shall be included in the weight report (see Section 5.2). It is possible that the tilt of the lifted equipment is different in air and water. The designer shall determine whether the attitude of the lifted equipment is critical in either.
- 11.3.6 **Dynamic motion.** During the lift off the vessel/barge deck, through the splash zone and lowering through the upper sections of the water column the lifted equipment should be controlled using tugger or control lines. Set down loads shall be calculated based on the heave response of the lifted equipment in the water column, and the set down speeds established to limit impact loads on the equipment, host structures and/or guidance systems.
- 11.3.7 **Natural frequency.** For deep-water installations the natural frequency changes as the lifted equipment is lowered through the water column. Installation analyses should be undertaken to determine if there is a possibility of resonance. When resonance is possible the effects shall be quantified and mitigating measures identified to overcome its effects.
- 11.3.8 **Phasing.** For deep-water lowering operations the effects of phasing between the boom tip and the lifted object shall be investigated by analysis and limiting seastates identified. Special lifting slings, ropes and devices may be needed to limit these effects.
- 11.3.9 **Mid-water load transfer** is possible for any item, but this shall be carried out at a depth so that any lateral load on a rigging or crane system is minimised.
- 11.3.10 **Dropped objects.** Consideration shall be given to the safety of existing equipment and pipelines on the seabed during overboard lifting operations. Where appropriate, the structure should be lifted overboard and lowered a safe horizontal distance from any existing subsea equipment and pipelines and then moved into the final position at a suitable height above the sea bed.

11.4 DEPLOYMENT SYSTEM

- 11.4.1 The deployment system shall be capable of raising the structure off the vessel deck and over-boarding through the splash zone and to its final location with sufficient motion control.
- 11.4.2 The deployment system shall also be capable of a complete reversal in the event of any unforeseen event taking place.
- 11.4.3 A dual deployment system, where the structure is lifted off the deck, lowered through the splash zone to a suitable depth below the sea surface using the crane and then transferred to a deployment winch for the remaining lowering to the seabed, is acceptable.
- 11.4.4 The crane shall be certified for its intended function and load-radius curves shall be provided. In certain cases this may include different vessel roll angles or DAF values.
- 11.4.5 If a deployment winch is chosen for lowering a structure to the seabed it must satisfy the same requirements as a crane, in terms of redundancy, FMEA, capacity and DAF.
- 11.4.6 Load cells on deployment winches shall be calibrated annually against certified load cells and load test certificates made available to the attending surveyor.

11.5 POSITIONING AND LANDING

- 11.5.1 It is suggested that the set down speed is limited to a maximum value of 0.5m/sec and impact loads to no more than 3% of the submerged weight; however the manufacturer's requirements shall be followed when they are more onerous.
- 11.5.2 Surface guide-wires can be used to control the position and orientation of the lifted equipment as it is lowered into position. The guide-wires can either be secured to the installation vessel or be attached to sub-surface buoyancy elements. Surface buoys can be used but their interference with other down lines and umbilicals shall be considered in their design, deployment and operation.
- 11.5.3 Visual (ROV) monitoring of the touchdown of the lifted equipment onto the seabed or the host structure is required. ROVs can also be used to control position / orientation / landing of light structures subject to the ROV capacity on location.

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- 11.5.4 Subsea guide-wires attached to subsea buoyancy can be used in deep water and are connected by the ROV after over-boarding of the lifted equipment.
- 11.5.5 Subsea rigging release systems shall be designed so that the crane vessel can quickly be disconnected from the lifted equipment. There should also be contingency systems in place to re-connect the rigging if retrieval is required. Hydraulic shackles and ROV release systems shall be certified and function tested.
- 11.5.6 The seabed condition should be considered if wet parking is planned. The consideration shall include bearing capacity and seabed inclination. It is possible that seabed preparation can be needed.

11.6 ROV SYSTEMS

- 11.6.1 ROV systems and tooling shall be selected based on the environmental conditions that are to be expected at the worksite during the planned and contingency intervention / observation tasks.
- 11.6.2 ROV-dependent operations shall be carried out only with vessels equipped with 2 or more ROV's. ROV thrust capacity should be 30% more than that required for the current speeds given in the site specific environmental reports.
- 11.6.3 ROV tooling shall be provided with sufficient spares and back-up tooling to allow the work to proceed with minimum delay.
- 11.6.4 It is recommended that a tether management system (TMS) be used in deepwater sites to ease the deployment of the ROV to the worksite. The tether shall be of sufficient length to allow the ROV to get from the TMS to the worksite.
- 11.6.5 Grab bars to aid ROV positioning for manipulative or observation tasks should be provided where critical path ROV operations are planned.
- 11.6.6 Once installed the launch and recovery system (LARS) shall be load tested to a factor of safety of 2.0.

11.7 TESTING

- 11.7.1 System Integration Testing shall be carried out onshore to prove that the integration of all components and tooling can be achieved. This may involve the manufacture of mock-ups. If mock-ups are used, great care shall be taken to ensure that the mock-ups replicate the actual item.
- 11.7.2 Dry tests and FAT should be carried out for critical and complex systems, the failure of which would result in significant and un-acceptable schedule delay.
- 11.7.3 Wet testing shall be considered for the actual ROV system to be used.

11.8 SUCTION PILES & FOUNDATIONS

- 11.8.1 A dynamic lift lowering analysis shall be carried out to determine the dynamics in the lifting system under the design installation seastates. The installation analyses should demonstrate that for the design sea states the integrity of the pile will be maintained for all stages of installation including the effects of slamming when lowering through the splash zone.
- 11.8.2 Heading control using subsea or surface guide wires shall be provided where the pile heading is critical. Pre-installed mooring lines and / or chains can be used to assist in heading control.
- 11.8.3 A system to monitor the verticality of the suction pile during installation shall be provided. This can be a visual system such as a calibrated bullseye.
- 11.8.4 The pile position shall be determined using a calibrated subsea positioning array.
- 11.8.5 The self-penetration of the suction pile should be estimated prior to the installation operations and compared to the actual value. This can provide information on the actual soil strength. If a large variation occurs the pile design should be re-evaluated.
- 11.8.6 Penetration indicators ("draught" marks) on suction piles shall be used to allow the initial self-penetration and final penetration of the pile to be determined visually.
- 11.8.7 On bottom stability of the pile prior to activating suction systems shall be calculated. For this aspect, short stubby piles are preferable to tall slender suction piles.

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- 11.8.8 ROV mounted suction pumps should be integrated into the host ROV and have sufficient flow to evacuate the water from inside the pile without exceeding the limiting differential pressures imposed by the capacity of the pile. The pump skid flow shall be reversible to allow retrieval of the pile if needed.
- 11.8.9 Independent pump skids can also be used and in some cases these can also have heading monitoring systems, altimeters and gyro compasses integrated into their control systems. Pump skid flow shall be reversible.
- 11.8.10 Integration testing of the mating flange for the pump skid should be performed prior to deployment.
- 11.8.11 Pile hydrostatic collapse and piping of external seawater through the soil shall be prevented by ensuring that the differential pressure between the inside and outside of the pile is kept within limits. Pump curves shall be provided and used.
- 11.8.12 Suction piles may have to be transported, over-boarded or lifted offshore from a barge horizontally due to limitations of the installation vessel, available crane hook height and to limit working at height offshore to connect the crane hook with installation rigging. In such conditions suction piles need to be upended before lowering to the seabed.
- 11.8.13 Lowering a suction pile horizontally through the splash zone can generate additional hydrodynamic loading due to increased surface area of the pile being presented to wave and current. These possibilities should be investigated during an installation analysis and adequate mitigating measures should be put into the operational procedures.
- 11.8.14 Upending should be performed by the gradual transfer of load from horizontal transfer rigging to installation rigging at a water depth where there are no possibilities of the suction pile and rigging system reaching resonance. Analyses should be performed to confirm such possibilities do not exist. The water depth chosen should be away from the influence of the splash zone to limit hydrodynamic loads acting on the pile. Once upending is completed the transfer rigging should be slackened and removed. Installation of the suction pile continues with the pile in a vertical orientation with the heading of the suction pile ready for landing on the seabed.

11.9 DRIVEN ANCHOR PILES

- 11.9.1 Unless dynamic effects are shown to be insignificant, a dynamic lift lowering analysis shall be carried out to determine the dynamics in the lifting system under the design installation seastates. Installation analyses should be used to demonstrate that for the design seastates the integrity of the pile will be maintained for all stages of installation including the effects of slamming when lowering through splash zone.
- 11.9.2 Subsea pile guide frames should be used to ensure that the anchor pile is driven with the required verticality, position and heading.
- 11.9.3 Pile driving loads in the pile guide frame shall be calculated to ensure that the frame has adequate capacity under all loading conditions including accidental and lateral loads.
- 11.9.4 A system to monitor the verticality of the anchor pile during driving should be provided. This can be a visual system such as a calibrated bullseye indicator.
- 11.9.5 On bottom stability of the guide frame shall be calculated based on the maximum inclination due to the seabed bathymetry with and without the pile hammer string. The mudmat shall be designed to ensure stability of the guide-frame and verticality of the anchor pile and pile hammer combination.
- 11.9.6 Pile driveability analyses shall be provided to ensure that the pile can be driven to the target penetration with the chosen pile hammer.
- 11.9.7 The self-penetration of the pile should be estimated prior to the installation operations and compared to the actual value. This can provide information on the actual soil strength. If a large variation occurs the pile design should be re-evaluated.

11.10 JUMPERS AND TIE-IN SPOOLS

- 11.10.1 Subsea jumpers and spools shall be arranged in such a way that they are appropriately supported during installation. In cases where the spool or jumper is significantly lighter than or buoyant relative to the supporting spreader bar, consideration should be given to using a “strongback” type spreader bar (strapped to the spool), as opposed to a “floating” spreader bar.

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- 11.10.2 A dynamic lift lowering analysis shall be carried out to determine the dynamics in the lifting system under the design installation seastates. The installation analyses should be used to demonstrate that in the design seastates the integrity of the structure will be maintained for all stages of installation including the effects of slamming when lowering through splash zone.
- 11.10.3 Spreader bars shall be of free flooding design where ever possible. In situations where the submerged mass of the spreader bar is critical, the design of the spreader bar shall have a factor safety of 1.5 against hydrostatic collapse at its maximum water depth.
- 11.10.4 The submerged and in-air weight of the spool shall be available from a weight report and include spool buoyancy.
- 11.10.5 The submerged and in-air CoG shall be included in weight reports in order that the correct value of tilt can be determined for installation and set-down.
- 11.10.6 Indeterminate rigging systems shall be designed accounting for actual sling lengths and the catenary effects of the rigging component self-weight. Means to adjust the sling lengths or geometry such as turnbuckles or moveable attachments can be used so that skew effects and loss of individual sling tension can be minimised.
- 11.10.7 Lift analyses for statically indeterminate rigging systems shall be carried out in order to quantify the load in each sling and show that the stress in the spool is within allowable limits. The maximum loaded sling(s) in the lifting arrangement should be removed in the analysis model to demonstrate that the rigging arrangement can support the spool without the spool being overloaded. In the absence of this analysis an SKL=1.75 shall be used for statically indeterminate rigging systems.
- 11.10.8 Trial lifting of spools and/or jumpers shall be carried out to verify the rigging geometry prior to load-out in order to:
- obtain the correct tilt angle when the inclination is critical or there is a significant difference between the in-air and submerged condition
 - verify that all slings are in tension for spool lifts.
- If the trial lift reveals that a sling is slack, the sling length shall be adjusted and the test lift repeated. Small movements in the positions of slings on the spool can often be used to even out the loads in the slings.
- 11.10.9 All soft slings that are choked around the spool shall be designed in such a way that their release can be made effectively.
- 11.10.10 When lifting off the deck of a transportation barge and lowering through the splash zone adequate clearance (at least 3m) should be maintained between the spool and the installation vessel.
- 11.11 RIGID PIPE RISER INSTALLATION**
- 11.11.1 This section covers rigid pipe that is to be attached to the exposed parts of a fixed installation resting on the seabed, e.g. a steel jacket structure, where the riser is installed in one or more sections (or spools) or, in rare cases, attached to the pipeline.
- 11.11.2 Riser spools can vary significantly in length and size. Short spools will be accommodated and seafastened on deck by lashings or weldments and clamps as appropriate. Large spools may have to be cantilevered or carried on over-side seafastenings welded to the installation vessel or to a transport barge. Transportation and seafastening of spools over-side of the vessel is covered in 0030/ND, Ref. [5]. Lifting of all but the shortest spools is covered in Section 11.10.
- 11.11.3 Spools with a length to diameter ratio of less than 10-15 need not normally be subject to dynamic analysis for the lift and instead may be designed simply with a DAF of 2.0 on the nominal weight from the piping and its lifting aids.
- 11.11.4 Prior to riser spool lifting, riser support clamps will be installed and adjusted to the geometry of the riser using taut wires or laser geometry to line them up. As the geometry can be quite complex, the fabricated spool must match the geometry through the clamps within the design tolerance allowed. The nominal design height from the lowest clamp to the seabed needs to be checked in the field for consistency with the design tolerance.

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- 11.11.5 Spools will be lifted into place, secured and clamped in a sequence suited to the platform and the joining method. Note that codes and standards limit the angular offset and the hi-lo at piping butt joints. At flanged joints it is not acceptable to attempt to close flanges which are visibly misaligned either torsionally, radially or subtending an angle between their axes.
- 11.11.6 Spool(s) from the lowest clamp to the seabed touchdown, and onward to the joint with the pipeline, are normally designed to flex elastically to accommodate platform settlement and pipeline expansion. Consequently, it is also important that these fit in accordance with the design dimensions and tolerances.
- 11.11.7 Lift rigging and lifting aids should normally comply with the requirements of Sections 5 and 11.3. Lifting points should not be welded directly to the pipe. Note that for long spools there may be a need to upend sections to assemble the spool in the field. The lifting design should cover all phases of the spool lifting and fabrication operations.
- 11.11.8 The tail of the riser and the pipeline will be moved into alignment for connection. Usually this entails lateral movement of the seabed pipeline on H-frames in controlled steps. If welded, the tail of the riser and the end of the pipeline will be lifted to welding height in a habitat or cofferdam. If flanged, the tail of the riser will be raised a little off the seabed, often on airbags or on an H-frame, to allow access to the flange for connection and tightening. All pipeline and riser movements should be analysed for load and stress to confirm equipment loadings and that pipe stresses are acceptable and to optimise the locations of H-frames, etc, for pipe level and angle at the connection.
- 11.11.9 Once connected, if flanged and already hydrotested, the line will be leak tested; otherwise it will be NDT tested and hydrotested. In some particular circumstances a so called "Golden Weld" will be permitted which is not required to be strength tested by hydrotest. Codes and Regulators normally only permit this where the hydrotest would expose other parts of the system which cannot be isolated to unacceptably large stresses or it is not practical to flood and test the pipeline to achieve a hydrotest. In lieu of hydrotesting, stringent additional NDT is required.
- 11.11.10 Leak tests are normally required at or above 1.1 times MAOP (Maximum Allowable Operating Pressure) for 4 hours, whereas hydrotest is normally for 24 hours at or above 1.5 times pipeline design pressure. The test specification should govern and define the acceptance criteria for any unaccounted pressure loss.
- 11.12 SUBSEA STORAGE TANKS**
- 11.12.1 Requirements for towage or transportation to the installation site are covered in 0030/ND, Ref. [5].
- 11.12.2 Any major compartment, whose buoyancy is required for intact or damaged stability, shall be pressurised to a minimum of 0.34 bar (5 psi). Compartment pressures shall be monitored for a period of three days prior to sailaway, immediately prior to sailaway and immediately upon arrival at the installation site. The method of monitoring the pressures shall be stated.
- 11.12.3 Careful attention should be given to the design of the primary and secondary flooding and venting systems to minimise the risk of premature flooding of the main and trimming tanks. The damage cases considered shall include the effects of any valve failing to open or close (or stay open or closed) at any relevant stage.
- 11.12.4 A dynamic lift analysis shall be carried out to identify the limiting seastates for lowering and hence establish the associated DAFs (Dynamic Amplification Factors).
- 11.12.5 Ideally the lowering system should be reversible though this may not be feasible for deballasting through the splash zone. The system should be designed to allow the storage tank to be repositioned on the seabed if the initial position is out of tolerance.
- 11.12.6 The floating stability and reserve buoyancy of the storage tank shall be analysed for the floating phases and from submergence through the splash zone to landing on the seabed. The effects of any cranes, winches, floating buoys and/or heave compensation systems used to control the descent should be considered in the analyses. The effects of the loss of any one lowering line or flooding of any one compartment at any stage shall be determined. If any such loss or flooding would lead to loss of the tank then suitable mitigation plans should be presented to GL Noble Denton for approval and be the subject of risk assessments to show that the risks are acceptable.

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- 11.12.7 Tug movements shall be given careful consideration to reduce the probability of tank damage during tow or operations afloat.
- 11.12.8 Initial ballasting of the storage tank will typically be carried out with the tank held in position by tugs about 50 to 100m away from the installation vessel. This distance selected should be sufficient to avoid contact but close enough for monitoring.

12 OPERATIONS AND PRACTICAL CONSIDERATIONS

12.1 ORGANISATION, PLANNING AND DOCUMENTATION

12.1.1 See Section 6 of 0001/ND, Ref. [1] for information on Organisation, Planning and Documentation.

12.1.2 Risk assessments and HAZOP/HAZID studies shall be carried out by the Contractor in the presence of the Client, GL Noble Denton and actual Contractor's operational personnel. These studies shall be completed at an early stage so that the findings can be incorporated into the operational procedures.

12.1.3 Operating manuals shall be prepared and agreed by all relevant parties in advance and include:

- management and communication systems including organograms and roles and responsibilities
- relevant drawings, specifications and calculations
- operating procedures including arrangements for weather forecasting, control, manoeuvring and mooring of barges and/or other craft alongside the crane vessel
- contingency procedures and emergency plans
- other items covered in Sections 5 and 6 of 0001/ND, Ref. [1]

12.2 SAFETY

12.2.1 See Section 5 of 0001/ND, Ref. [1] for information on Health, Safety and Environment.

12.3 WEATHER-RESTRICTED OPERATIONS AND WEATHER FORECASTS

12.3.1 Section 7.3 of 0001/ND, Ref. [1], "General Guidelines for Marine Projects" applies for all weather-restricted operations.

12.3.2 Section 7.4 of 0001/ND, Ref. [1], "General Guidelines for Marine Projects" applies to weather forecasts.

12.3.3 In field monitoring of waves (height, direction and period) should be considered to enhance the forecasts for each specific lift operation where a Certificate of Approval is required.

12.3.4 In field monitoring of currents (speed and direction) for subsea lifts in areas where it is known that high currents exist in the water column should be considered to enhance the forecasts where a Certificate of Approval is required.

12.4 ENVIRONMENTAL DESIGN CRITERIA

12.4.1 Section 7.2 of 0001/ND, Ref. [1], "General Guidelines for Marine Projects" applies to metocean criteria.

12.4.2 Seasonal current speeds with water depth variations (part from in shallow water), wave and wind data shall be provided for the field location, in the form of monthly or seasonal scatter diagrams.

12.4.3 The design conditions for installation shall be agreed with GL Noble Denton at as early a stage as possible.

12.5 SURVEY AND POSITIONING

12.5.1 A geophysical and geotechnical survey of sufficient resolution shall be carried out during the detailed design phase to determine the nature and the bathymetry of the seabed. This is to include soil characteristics and stability if required.

12.5.2 Comprehensive pre-installation site survey reports shall be issued (local structures, seabed bathymetry, obstacles, pipelines etc.).

12.5.3 Surface positioning for installation shall be achieved using a calibrated GPS or DGPS system. There shall be back-up reference systems available in the event that the primary reference system fails.

12.5.4 If a structure has to be positioned subsea in a location where there is no seabed infrastructure, a pre-calibrated LBL or USBL array may be required. Transponders shall be deployed and surveyed in on the installed structure so that it interfaces with the array in terms of position and heading. Survey / structure acoustic systems shall be designed and documented such that conflicts with DP References, ROV transponders and divers are avoided.

12.6 VESSEL MOTIONS

12.6.1 Crane vessel motions should be monitored in the period prior to the lift to confirm that the dynamic behaviour is acceptable taking into account:

- The weight and size of the lifted object,
- The clearances for lifting off the transport barge,
- The hoisting speed,
- The clearances for installation and
- The installation tolerances.

12.6.2 Transport barge motions should be similarly monitored prior to the start of the lift. The change in attitude of the transport barge when the weight is removed should be taken into account.

12.6.3 When setting down on a floating structure, the set down procedure shall show how the lifted load will affect the draft and trim of the floating structure. The allowable range of lowering speeds shall be determined to avoid snatch loads, lift off or excessive motions of the floating structure.

12.7 SAFE ACCESS

12.7.1 Adequate and safe access and working platforms should be provided for connection of slings, particularly where connection or disconnection is required offshore or underwater.

12.8 LOOSE EQUIPMENT

12.8.1 All loose equipment, machinery, pipework and scaffolding shall be secured against movement during the lift, and the weights and positions allowed for in the gross weight.

12.9 SEAFASTENING REMOVAL

12.9.1 Seafastening on the transport barge should be designed:

- To minimise offshore cutting
- To provide restraint after cutting
- To allow lift off without fouling.

12.9.2 All cut lines should be clearly marked. Where a 2-stage lift is planned - e.g. barge to lift vessel, then lift vessel to final position, involving 2 sets of cut lines, these should preferably be in different colours.

12.9.3 Where clashes with the lifted structure might occur during the lift, the primary mitigation is to ensure that all secondary steel that has the potential for clashing with the structure is marked for pre-lift removal; the selection criteria should be as in Section 9.2.

12.9.4 If the transport vessel is to be moved as part of a lifting operation it is important that all constraints are documented as part of the lifting procedure to prevent clashes, e.g. the barge shall be removed in a specific direction only. In all cases the constraints on the operation (the lift and associated post-lift vessel movements) should be clearly documented in the lift procedure.

12.9.5 Adequate equipment must be available on the transport barge, including as appropriate:

- Burning sets
- Tuggers and lifting gear
- Means of securing loose seafastening material
- Lighting for night operations
- Safety equipment for personnel.
- Safe access to and from the transport barge.

12.10 SLINGS & SHACKLES

- | | | |
|----------|--|----|
| 12.10.1 | The sling laydown arrangement shall show that: | |
| | <ul style="list-style-type: none"> a. The slinging arrangement is in accordance with acceptable good practice. b. Large diameter slings and grommets shall be painted with a white line along the length to monitor twisting during handling and laydown. c. The sling lengths are matched as accurately as possible, unless the rigging arrangement is deliberately non-symmetrical to take account of centre of gravity offset, in which case matched pairs of slings should normally be used. Where minor mismatch in sling length exists, the slings should be arranged to minimise skew loads. d. The slings are adequately secured against barge motions and protected from abrasion against sharp edges and adjacent structure. (Where sacrificial lashings are used, see Section 6.7.4 of 0028/ND, Ref. [3] and GPR.03 in Appendix I of 0030/ND, Ref. [5]) e. The slings will not foul obstructions such as walkways and handrails when lifted, and any unavoidable obstructions are properly protected. f. The slings will not kink when lifted. g. After the lift the slings (and spreaders if used) can be safely laid down again, without damage. h. In the event that a single sling attached to a single lift point is planned, it should be doubled to prevent the sling unwinding under load. | 10 |
| 12.10.2 | Slings with hand spliced terminations must be prevented from rotation. | |
| 12.10.3 | No bending is allowed at or close to a termination. | |
| 12.10.4 | It is permissible to shackle slings together end-to-end to increase the length. However, slings of opposite lay should never be connected together. | |
| 12.10.5 | It is permissible to increase the length of a sling by inserting an extra shackle (but not a wide body shackle because it should not be connected bow-to-bow) or specifically designed link plates. Any shackle to shackle connections should be bow-to-bow, not pin-to-pin or pin-to-bow (unless specifically allowed by shackle manufacturers) so that shackles remain centred under the load and also during the load take-up. | 10 |
| 12.10.6 | Slings and grommets should be manufactured and inspected in accordance with the IMCA Guidance on Cable laid slings and grommets, Ref. [7], or similar acceptable standard. A thorough examination shall be carried out as required by that document for all rigging components whether new or existing. | |
| 12.10.7 | Shackles shall be manufactured by, and covered by a certificate from, an industry-recognised manufacturer. If this certificate is more than 2 years old then there shall also be a MPI /UT test certificate less than 2 years old. If this test certificate is more than 6 months old then there shall also be a report of an inspection by a competent person within the previous 6 months. The lift shall be executed within the date validity (if any) of the shackle certificate. | 10 |
| 12.10.8 | When shackles are used on a regular basis, a visual inspection shall be carried out by a competent person before any lift when the utilisation of the shackle will be more than 80% of the SWL. | |
| 12.10.9 | Where 9-part slings are proposed for use in a lifting system, certification of these slings shall be given special consideration. | |
| 12.10.10 | Where an existing sling has been used doubled and this sling shows a permanent kink, it shall not be used in a single configuration. | |
| 12.10.11 | Where spreader bars or spreader frames are used in a lifting system, there shall either be a load test certificate provided indicating the SWL or WLL, tested in accordance with GL Guidelines, Ref. [11], or an as-built dossier provided with data as listed in Section A.3.2. Where a single point of failure in the lift system does not cause the loss of the lifted item, contingency procedures/plans should, where possible, be in place to allow it to be safely recovered. | 10 |

12.11 LIFTING TOOLS

12.11.1 Where lifting tools are used as part of a lifting arrangement, the maximum loads imposed on such tools shall not exceed the stated certified WLL for the tool. The hydraulic system (if used) should be of a fail-safe nature, such that in the event of loss of hydraulic power, the tool will remain in fail-safe mode. Test certificates shall be issued or endorsed by a body approved by an IACS member for the certification of this type of equipment.

12.11.2 **External and internal hydraulic lifting tools** shall always have:

- a. remote monitoring system close to the crane driver's cab
- b. a pressure gauge (or indicator) in the system showing when the tool is closed or open
- c. a duplicate pressure gauge (or indicator), as close as safely possible to the tool to avoid influences in pressure reading
- d. a method to release the tool in the event of hydraulic system failure.

12.11.3 **Automatic lifting tools** shall have systems in place to control the stress in the lifted item in order to prevent excessive local stress for sensitive items. Redundant mechanical systems must be in place in case of power loss.

12.12 COLOUR CODING

12.12.1 Consideration should be given to colour code the rigging, i.e. one colour per rigging set to each lift /upend point. This is good practice to avoid mix up, especially when the slings and connectors are pre-checked "loosely" on the ground or deck before attaching to the lift points.

10

REFERENCES

- [1] GL Noble Denton Report 0001/ND - General Guidelines for Marine Projects
- [2] GL Noble Denton Report 0013/ND - Guidelines for Load-Outs.
- [3] GL Noble Denton Report 0028/ND - Guidelines for Steel Jacket Transportation & Installation.
- [4] GL Noble Denton Report 0029/ND - Guidelines for Submarine Pipeline Installation
- [5] GL Noble Denton Report 0030/ND - Guidelines for Marine Transportations
- [6] GL Noble Denton Report 0032/ND - Guidelines for Moorings
- [7] International Marine Contractors Association - Guidance on the Use of Cable Laid Slings and Grommets - IMCA M 179 August 2005.
- [8] ISO International Standard ISO 19901-5:2003 – Petroleum and natural gas industries – specific requirements for offshore structures – Part 5: Weight control during engineering and construction.
- [9] ISO International Standard ISO 2408:2004 - Steel wire ropes for General Purposes - Minimum Requirements
- [10] ISO International Standard ISO 7531:1987 - Wire Rope slings for General Purposes - Characteristics and Specifications.
- [11] Germanischer Lloyd - Guidelines for the Construction and Survey of Lifting Appliances (1992 Edition)

All GL Noble Denton Guidelines can be downloaded from http://www.gl-nobledenton.com/en/rules_guidelines.php

APPENDIX A - INFORMATION REQUIRED FOR APPROVAL

A.1. GENERAL INFORMATION REQUIRED

- A.1.1 Where approval is required, a package shall be submitted to GL Noble Denton for review, consisting of:
- Structural analysis report for structure to be lifted, including lift points and spreaders, as in Sections 7 and A.2.
 - Rigging arrangement package as in Section A.3.
 - Details of the lifting vessel, cranes and mooring /DP systems as in Section A.3.3.
 - The management structure, risk assessments and marine manuals /procedures as in Section 12.
 - A site survey of the installation area covering the full area of any anchor pattern, carried out not more than 4 weeks before the start of installation, shall be provided to verify the location of all subsea infrastructure, debris and obstructions.

A.2. THE STRUCTURE TO BE LIFTED

- A.2.1 Calculations shall be presented for the structure to be lifted, demonstrating its capacity to withstand, without overstress, the loads imposed by the lift operation, with the load and safety factors stated in Section 5, and the loadcases discussed in Section 7. The calculation package shall present, as a minimum:
- Plans, elevations and sections showing main structural members
 - The structural model. This should account for the proposed lifting geometry, including any offset of the lift points
 - The weight and centre of gravity, including justification of weight and centre of gravity, by Weight Control Report or weighing report, as in Section 5.2. For subsea lifts the weight report shall include the submerged weight and centres of gravity and buoyancy.
 - For partly immersed and subsea lifts the requirements of Section 11.2 and 11.3 shall be addressed.
 - For subsea lifting or lowering the additional information covered in Sections 11.2.8 and 11.3.
 - The steel grades and properties
 - The loadcases imposed
 - The Codes used
 - A tabulation of member and joint Unity Checks greater than 0.8
 - Justification or proposal for redesign, for any members with a Unity Check in excess of 1.0.
- A.2.2 An analysis or equivalent justification shall be presented for all lift points, including padeyes, padears and trunnions, to demonstrate that each lift point, and its attachment into the structure, is adequate for the loads and factors set out in Sections 5 and 7.
- A.2.3 A similar analysis shall be presented for spreader bars, beams and frames.
- A.2.4 Confirmation shall be presented, from a Certifying Authority, Classification Society or similar, that the structure including the lift points and their attachments has been constructed in accordance with the drawings and specifications.

A.3. RIGGING ARRANGEMENTS

- A.3.1 Documentation shall be presented including:
- a. The proposed rigging geometry showing:
 - Dimensions of the structure,
 - Centre of gravity position,
 - Lift points,
 - Crane hook,
 - Sling lengths and angles, including shackle dimensions and "lost" length around hook and trunnions.
 - b. A computation of the sling and shackle loads and required breaking loads, taking into account the factors set out in Section 5.
 - c. A list of actual slings and shackles proposed, tabulating:
 - Position on structure
 - Sling/shackle identification number
 - Sling length and diameter
 - Rigging utilisation factor summaries
 - CSBL, CRBL for slings or CGBL for grommets,
 - SWL or WLL for shackles
 - Construction
 - Direction of lay
 - Wire grade and wire type (bright or galvanised).
 - Copies of inspection/test Certificates for all rigging components. These certificates shall be issued or endorsed by a body approved by an IACS member for the certification of that type of equipment.
- A.3.2 Where spreader bars or spreader frames are not load tested (as in Section 12.10.11) an as-built fabrication dossier shall be provided listing the following minimum information:
- a. Material certificates (3rd party endorsed),
 - b. Welding consumables certificates,
 - c. Weld procedures,
 - d. NDT procedures,
 - e. Welders and NDT operatives qualifications,
 - f. Inspection and Test Plan (ITP) listing Hold Monitor and Witness points,
 - g. 3rd party fabrication release note,
 - h. Technical queries /concession requests,
 - i. As-built drawings,
 - j. Design report.
- A.3.3 When subsea lifting or lowering is involved, additional documentation should be provided to show that the topics covered in Section 11.3 to 11.12, if applicable, are covered.

A.4. THE INSTALLATION VESSEL

A.4.1 Information shall be submitted on the installation / crane vessel and any cranes or winches to be used. This shall include, as appropriate:

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- a. Vessel general arrangement drawings and specification including proposed operating and survival drafts.
- b. Documentation and certificates (see Section 6.6 of 0001/ND, Ref. [1]).
- c. Vessel station keeping procedures (see Section 13 of 0001/ND, Ref. [1]).
- d. DP system and documentation (as applicable) as in Section 6.5.
- e. Mooring system and anchors (as applicable) as in Section 6.6.
- f. Crane specification and operating curves (including where necessary the dynamic crane capacity / curve) and heave compensation).
- g. Qualification and certification of crane operators.
- h. Details of any ballasting operations required during the lift.
- i. For subsea lifting or lowering, details of:
 - Where necessary for the operation, ROV's & tooling (see Section 11.6);
 - Details of any separate winches & heave-compensation systems to be used (see Section 6.3).

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